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Advances in Hazard Management for Machine Learning based Safety-Critical Systems

Operating in Unpredictable Environments

by

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Thesis for the degree of Doctor of Philosophy

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University of Southampton

Abstract

Faculty of Engineering and Physical Sciences

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Doctor of Philosophy

Appendix H Unsafe Train Tracks Problem Case Study

by

Haider M. Al-Shareefy

This appendix accompanies the doctoral thesis entitled *Advances in Hazard Management for Machine Learning based Safety-Critical Systems Operating in Unpredictable Environments*.

Appendix L presents the safety case construction artefacts and consolidation evidence demonstrating alignment between the AIC Systems Approach and two established safety assurance frameworks: the Safety Case for Autonomous Systems (SACE) and the Assurance of Machine Learning for use in Autonomous Systems (AMLAS).

The appendix opens with a reusable STCoT Validation Report Template (§L.1), followed by its instantiation for two case studies: the AVOIDDS autonomous airliner dataset (§L.2) and the unsafe train tracks Eagle Drone system (§L.3). Each validation report documents the scope, artefact outputs, and recommended next steps produced by the AIC-based analysis.

The SACE consolidation (§L.4) maps AIC approach outputs, Stage 1 through Stage 4, to six SACE assurance stages, from Operating Context through Hazardous Failures Management, demonstrating that the AIC approach generates artefacts directly traceable to SACE's structured argumentation patterns. A summary consolidation table is provided in §L.4.9.

The AMLAS safety case argumentation patterns (§L.5) trace the Eagle Drone ML development pipeline through four AMLAS stages, Safety Assurance Scoping, ML Safety Requirements, Data Management, and Model Learning, providing auditable evidence that the proposed Black Swan-driven development approach satisfies recognised ML assurance guidance.

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Research Thesis: Declaration of Authorship

Print name: Haider Al-Shareefy

Title of thesis: Advances in Hazard Management for Machine Learning based Safety-Critical Systems
Operating in unpredictable environments

I declare that this thesis and the work presented in it are my own and has been generated by me as the result of my own original research.

I confirm that:

This work was done wholly or mainly while in candidature for a research degree at this University. Furthermore,

Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated. Furthermore,

Where I have consulted the published work of others, this is always clearly attributed. Furthermore,

Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work. Furthermore,

I have acknowledged all main sources of help. Furthermore,

Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself. Furthermore,

Signature: Date: 16/03/2025

Acknowledgements

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Appendix L Safety Case Construction and AIC consolidation with AMLAS/SACE approaches

Note: We often refer to the artefacts in Appendices H and I because they contain detailed tables and figures. However, each section in Appendices H and I is also associated with sections in Chapters 6, 7 and Appendix M.

L.1 STCoT Validation Report Template

This template provides a structured approach for validating Systems Engineering Chains of Thought (STCoTs) in a systematic and repeatable manner. It ensures that each STCoT follows a rigorous predictive thinking process, aligning with general systems principles and enabling a thorough exploration of emergent behaviours. The template can be adapted for any STCoT by defining specific elements and applying the methodology to various systems engineering contexts. For a full description of STCoTs, see section D.8. for full definition of STCoT methods. The following are the main sections of the report:

L.1.1 Purpose of the Report

- Clearly define the purpose of the validation report.
- Explain why the STCoT needs to be validated.

Example: This report validates the "AIC Perspective Shift STCoT," ensuring that its predictive thinking steps systematically uncover unforeseen scenarios that impact autonomous safety assurance.

L.1.2 Scope of Validation

- Define the boundaries of what is being validated.
- Specify whether the validation applies to a single STCoT, a set of STCoTs, or a broader system application.
- You may also include "Out of Scope" to summarise what would be outside the scope of validation.

Example: The scope of this validation includes examining predictive thinking pipelines, general systems rules, and structured outputs to ensure alignment with the STCoT_1 principle.

L.1.3 STCoT Output Validation

L.1.3.1 Validation Criteria

- Define measurable criteria (pass/fail) to determine whether STCoT predictions are valid and valuable.
- Include qualitative and quantitative measures.
- Examine if the step completion criteria in the STCoT are preserved.

Example criteria:

- The predictions align with general systems axioms.
- The STCoT process reveals previously unrecognised emergent behaviours.
- The thinking pipelines yield multiple alternative scenarios.
- Step completion criteria are preserved.

L.1.3.2 Validation Results Table

Define the appropriate textual description of the validation test that needs to be checked for a given STCoT.

Table L.1 STCoT validation results table

Validation Step	Evaluation Criteria	Pass/Fail	Notes
Step 1	Alignment with general systems rules.	Pass	Predictions adhere to known system behaviours.
Step 2	Generation of unforeseen scenarios.	Pass	Identified new emergent behaviours.
Step 3	Applicability to real-world cases.	Pass	Scenario aligns with real-world system failures.
Step 4	Step completion criteria are preserved.	Pass	The output predictions preserve step completion criteria.

L.1.4 Summary and Next Steps

L.1.4.1 Summary of Findings

- Summarize the key takeaways from the STCoT validation.
- Highlight whether the STCoT successfully identified emergent scenarios and preserved general systems rules.

Example: The STCoT validation confirmed that predictive thinking pipelines effectively identify unforeseen scenarios in autonomous drone operations, revealing three unaccounted failure cases.

L.1.4.2 Recommended Actions

Outline the next steps for refining the STCoT or integrating it into system design.

Example:

- Incorporate STCoT predictions into risk mitigation planning.
- Expand STCoT pipelines to cover additional interaction types.
- Validate STCoT predictions through simulation or real-world testing.

L.1.5 Appendices (Optional)

Include supporting materials such as:

- Diagrams of the STCoT predictive steps.
- Simulation results validating STCoT predictions.
- Additional examples of general systems rules in use.

L.1.6 STCoT Validation Review Process

- Validation reports must follow the peer review process.
- The process starts with the naming of the architect who is responsible for generating the report.

- It must also include signatures of at least one more reviewer to affirm the validity of the process.

L.2 STCoT Validation Report for AVOIDDS case study¹

Below is a sample STCoT Validation Report drafted by the L.1 STCoT Validation Report Template. This example applies to the “Identification and Analysis of Unsafe Problematic Behaviours in Mid-Air Collision” STCoT, Table I.1, validating the correctness and usefulness of “Architect Assertion 1.1.4” about confusing and unsafe behaviours that may lead to collisions.

L.2.1 Purpose of the Report

This validation report evaluates the “Identification and Analysis of Unsafe Problematic Behaviours in Mid-Air Collision” STCoT, ensuring that its predictive thinking steps systematically uncover unforeseen or unsafe behaviours in the problem domain. It also confirms whether the STCoT product meets the expected completeness, consistency, and alignment with general systems rules and organisational needs.

- **STCoT Under Review:** “Unsafe Problematic Behaviours List Identification”
- **Architect Assertion 1.1.4:** Outlines two major problematic behaviours in mid-air collision contexts:
 1. Aircraft deviating unexpectedly from assigned flight paths
 2. By-passing aircraft moving erratically at high altitudes near crowded airspace.

L.2.2 Scope of Validation

This validation covers:

- Examination of how the STCoT identifies and describes confusing or unsafe behaviours in the domain of mid-air collision avoidance.
- Authentication that the **Predictive Thinking** steps are properly followed, i.e., that they adhere to “General rule A” and “General rule B” from your general systems rules.

¹ Associated with section 7.9 and table I.1

- Assess alignment with the Step Completion Criteria (1.1.3) to confirm that the STCoT product sufficiently describes the problem context and indicates undesirable impacts without prematurely jumping to design solutions.

Out of Scope:

- Detailed assessment of proposed solutions or mitigations for any of the identified unsafe behaviours.
- Authentication of other STCoTs beyond the “Identification and Analysis of Unsafe Behaviours” focus.
- Full end-to-end system safety or certification processes outside the immediate STCoT’s boundaries.

L.2.3 STCoT Output Validation

L.2.3.1 Validation Criteria

We define the following **pass/fail** criteria to determine whether the “Identification and Analysis of Unsafe Problematic Behaviours in Mid-Air Collision” STCoT output is valid and valuable:

1. Alignment with General Systems Rules

- The STCoT must illustrate that **General rule A** (unsafe behaviour is a type of confusing behaviour) and **General rule B** (confusing behaviours cause undesirable emergent outcomes) are correctly applied.

2. Relevance to the Stakeholder/Architect Sphere of Concern

- The problem statements must show that the described behaviours (e.g., erratic or unauthorised flight path deviations) fall within the domain’s sphere of concern (e.g., collision risk in congested airspace).

3. Identification of Realistic Undesirable Outcomes

- The STCoT must show how each unsafe or confusing behaviour leads to specific, undesirable consequences (e.g., near misses or actual collisions).

4. Exclusion of Solutions

- The deliverable should focus on analysing behaviours and not present solutions or mitigations (in line with the step’s instructions).

5. Step Completion Criteria (1.1.3) Preserved

- The STCoT must sufficiently describe the problem (the “Confusing Complex”) and the unsafe impact on some element B.

L.2.3.2 Validation Results Table

Below is an example table summarising the evaluation of each step or key aspect in the STCoT.

Validation Step	Evaluation Criteria	Pass/Fail	Notes
Step 1: Identify Problem Domain	Does the STCoT describe a mid-air collision scenario with enough clarity?	Pass	The problem domain is stated clearly: mid-air collisions. The scenario references “aircraft deviate from assigned path.”
Step 2: Apply General Rule A, B	Are unsafe/ confusing behaviours recognised as contradictory? Do they cause emergent risk?	Pass	Identified two primary behaviours: unexpected flight path deviation and erratic flight near crowded airspace, leading to collision risk.
Step 3: Problem-Focused, No Solutions	Does the text avoid introducing solutions?	Pass	The STCoT only defines problems: “Aircraft deviating” and “Moving erratically...” It does not mention new solutions or systems.
Step 4: Step Completion Criteria	Does an “undesirable outcome accompany each behaviour”?	Pass	The undesirable outcomes are clearly described as “Increased collision risk,” “ATC confusion,” and “reactive manoeuvres.”
Overall	Does the STCoT produce an acceptable description of the Confusing Complex?	Pass	The problem context is consistent, relevant, and no solutions overshadow the problem articulation.

L.2.4 Summary and Next Steps

L.2.4.1 Summary of Findings

- The STCoT effectively identifies critical unsafe behaviours (such as “unexpected flight path deviations” or “unauthorised, erratic flight in crowded space”).

- Each behaviour is framed as “confusing” and includes specific undesirable outcomes (e.g., collision risk, unpredictability for other pilots).
- The content adheres to General Rule A and General Rule B, explaining how each contradictory behaviour escalates collision likelihood and thus is within the “sphere of concern.”
- No solutions appear in the text, thus fulfilling the requirement to remain problem-focused.

All these observations indicate that the STCoT output meets the objectives set out in the instructions for “Identification and Analysis of Unsafe Problematic Behaviours.” Hence, the STCoT is validated as meeting the acceptance criteria for this stage.

L.2.4.2 Recommended Actions

To refine future STCoT steps or expansions, the following actions are suggested:

1. Feed into Next Phase

- Use these validated unsafe behaviours to inform the following process steps:
 - Step 1.2) Generate a descriptive image that visualises the unsafe behaviour

L.2.5 Appendices (Optional)

If needed, include:

- Minutes of problem articulation reviews with the customer that support this STCoT's finding.
- **Illustrative diagrams** or references to concept sketches of the flight paths, relevant logs of near-misses, or any additional data compiled during the problem articulation.

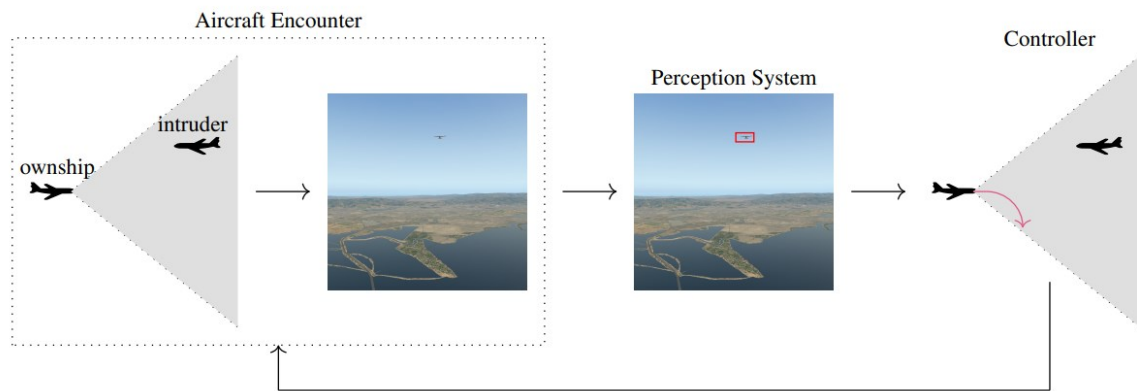


Figure L.1 Diagram taken from [1]

L.2.6 STCoT Validation Review Process

- **Report Author/Architect:** Haider Al-Shareefy [University of Southampton], responsible for generating the STCoT validation content.
- **Reviewer 1:** Prof Michael Butler [University of Southampton], validating alignment with General Rules D, E, F.
- **Reviewer 2:** Dr Son Hoang [University of Southampton], verifying coverage of elements and clarity of PrimePs.
- **Reviewer 3:** Prof Hamid Asgari [Thales], performs customer acceptance of validation report output.
- **Sign-off:** The above signatories affirm that the **STCoT validation** process was carried out thoroughly, and the results are valid as of this report date.

By signing below, the reviewers confirm that **I.4.1.1 Step 1.1) Identify a list of unsafe or confusing behaviours** of the STCoT is validated as meeting the defined criteria.

Signatures:

Signatures:

- Architect: _____ (Date: _____)
- Reviewer 1: _____ (Date: _____)
- Reviewer 2: _____ (Date: _____)
- Reviewer 3: _____ (Date: _____)

End of STCoT Validation Report

L.3 STCoT Validation Report for Unsafe Train Tracks case study²

L.3.1 Purpose of the Report

This report aims to validate the STCoT step “H.4.1.4 Step 1.4) Define the Supra-Complexes and Their PrimePs” ensuring that its predictive thinking processes and outputs adhere to the established general systems rules and follow the method’s intended objective.

The STCoT step under validation identifies two supra-complexes within the train tracks problem domain, Train Network and Adversarial Scheme, each associated with a distinct Prime Purpose (PrimeP). Validating this STCoT output is crucial to confirm that the methodology correctly organises the system elements into supra-complexes with consistent, clearly stated PrimePs, thereby guiding subsequent design and solution development.

L.3.2 Scope of Validation

- **Focus:** The validation is limited to the single STCoT step “Define the supra-complexes and their PrimePs” as documented in Table H.4 and summarised in Architect Assertion 1.4.6.
- **Boundaries:** The assessment excludes detailed downstream solutions, mitigations, or expansions of these supra-complexes beyond verifying that the STCoT step (1) correctly applies the relevant general system rules (D, E, F) and (2) satisfies the completion criteria (1.4.5).
- **Related References:** This validation specifically examines integrating with the “General Systems Rules” (especially D, E, F) and aligning with the AIC-based method for clarifying purpose and system scope.

L.3.3 STCoT Output Validation

L.3.3.1 Validation Criteria

The following criteria (Pass/Fail) are applied to judge whether the STCoT step output is acceptable:

1. Alignment with General System Rules

- **General Rule D:** Has the architect defined supra-complexes as “larger collections of complexes”?

² Associated with section 6.10 and table H.4

- **General Rule E:** Are the purpose statements consistent, stable, and clear across scenarios?
- **General Rule F:** Does each supra-complex have a well-defined Prime Purpose in the sense of a “Primary Purpose” that influences design?

2. Clarity of Supra-Complex Scope

- Each supra-complex must logically group relevant elements to achieve its chosen PrimeP.

3. Correctness and Completeness

- The final list of supra-complexes must encompass all visible elements.
- The step must not omit significant elements or add extraneous ones that do not fit the scenario.

4. Step Completion Criteria (1.4.5) Preservation

- The step’s deliverable is considered complete when at least one supra-complex is identified to represent each group of coexisting elements, and each is assigned a PrimeP.

5. Absence of Unwarranted Solutions

- This phase must remain descriptive about system structure and purpose rather than specifying a design or solution.

Validation Step	Evaluation Criteria	Pass/Fail	Notes
Step 1: Application of Gen. Rules D, E, F	Are supra-complexes formed properly (Rule D)? Are the defined PrimePs consistent (Rules E, F)?	Pass	Two supra-complexes, Train Network and Adversarial Scheme, are logically separated, each with a stable, domain-consistent PrimeP.
Step 2: Clarity of Supra-Complex Scope	Do these supra-complexes collectively include all identified elements?	Pass	The Train Network includes the train, fence, vegetation, and power lines. The Adversarial Scheme consists of the adversarial drone. Comprehensive coverage.
Step 3: Correctness and Completeness	Are the key elements (train, fence, drone, etc.) correctly allocated?	Pass	All visible elements are accounted for; each belongs to a supra-complex that logically groups them by purpose.

Step 4: Step Completion Criteria 1.4.5 Preserved	Is each supra-complex assigned a PrimeP?	Pass	Train Network → “transport people/goods safely” Adversarial Scheme → “disrupt train operations” Aligned with the method’s instructions.
Step 5: No Solutions Included	Does it avoid design solutions?	Pass	The final output remains at the level of problem domain structuring, no solutions or mitigations overshadow the definition process.

L.3.4 Summary and Next Steps

L.3.4.1 Summary of Findings

1. General Systems Rule Application

- The step properly operationalises Rule D by defining supra-complexes that unite sets of systems with shared overarching objectives.
- Rules E and F are upheld: each supra-complex’s PrimeP is well defined and consistent across potential scenarios.

2. Thoroughness in Coverage

- All major observable elements (train, tracks zone, vegetation, fence, power lines, drone) are included and correctly assigned to either the Train Network or Adversarial Scheme supra-complex.

3. Clarity for Future Stages

- The logic behind choosing these two supra-complexes clarifies how future design or risk analysis can proceed (e.g., focusing on how adversarial elements threaten the train network’s primary purpose).

Therefore, the STCoT step “Define the supra-complexes and their PrimePs” meets the validation criteria, with no unmet requirements or contradictions discovered.

L.3.5 Recommended Actions

1. Incorporate into Next Steps

- Use these supra-complex definitions to guide scenario-based hazard analyses or system design choices, especially in subsequent steps focusing on emergent interactions.

2. Validate Worst-Case Assumptions

- Validate or refine the assumption that the “Adversarial Scheme” might be more widespread than a single drone (i.e., a multi-drone orchestrated threat).

L.3.6 Appendices

- (If desired) Diagrams showing the two supra-complexes and their sub-elements.
- Additional references or notes on re-scope definitions, if the “Adversarial Scheme” requires broadening.

L.3.7 STCoT Validation Review Process

- **Report Author/Architect:** Haider Al-Shareefy [University of Southampton], responsible for generating the STCoT validation content.
- **Reviewer 1:** Prof Michael Butler [University of Southampton], validating alignment with General Rules D, E, F.
- **Reviewer 2:** Dr Son Hoang [University of Southampton], verifying coverage of elements and clarity of PrimePs.
- **Reviewer 3:** Prof Hamid Asgari [Thales], performs customer acceptance of validation report output.
- **Sign-off:** The above signatories affirm that the **STCoT validation** process was carried out thoroughly, and the results are valid as of this report date.

By signing below, the reviewers confirm that **H.4.1.4 Step 1.4) Define the supra-complexes and their PrimePs** of the STCoT is validated as meeting the defined criteria.

Signatures:

- Architect: _____ (Date: _____)
- Reviewer 1: _____ (Date: _____)
- Reviewer 2: _____ (Date: _____)
- Reviewer 3: _____ (Date: _____)

End of STCoT Validation Report

L.4 Consolidation with SACE Patterns

L.4.1 Stage 1: Operating Context Assurance

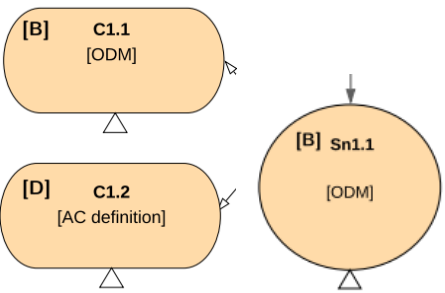
The first stage of SACE requires defining the AS's capabilities, validating its Operational Domain Model (ODM), and defining operating scenarios. The primary outcomes of this stage are:

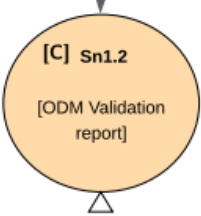
1. Autonomous Capabilities are specified.
2. AS's Operational Domain Model (ODM) are established and confirmed.
3. Operating Scenarios within the established ODM are identified and validated.
4. Development of Operating Context Assurance Argument.

AIC Systems Approach Processes involved to satisfy the objectives:

- Stage 1: Uncertainty Problem Articulation and Operational Environment Modelling
- Stage 2: Architect Intent and Autonomous Solution Needs Definition
- Stage 3A: HazTOPS and Ordered AIC-driven Autonomous System Requirements Development

L.4.1.1 Argument Pattern for Eagle Robot Operating Context Assurance

Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts
	The following process outputs satisfy the artefact's demonstration requirements: Table H.15 Architect High-Level Solution Prescription. • Sections 6.2, 7.2. Table H.35 Operational Design Definition for Eagle Robot Deployment in Train Track Zone

	Sections L.1 for STCoT validation template, sections H.12, L.2 for implementations.
	H.4.4.3 Step 4.3) Define the assumptions made about factors. Table J.8 unsafe train tracks problem domain assumptions.
	H.4.3.1 Step 3.1) Model detailed AIC interactions scenarios for the problem domain. Figure H.3 Modelling a complicated interaction n6. Table J.6 AIC problem domain scenarios definition
	H.4.3.2 Step 3.2) Predict the extended list of emergent AIC interactions scenarios. Table H.10 Complexity Field for n6 Interaction STCoT definition. Table J.7 AIC Extended Scenarios.
	H.6.1 Predictive Thinking Pipeline 1: Introducing Autonomous systems into Feed-forward complexity. Table H.18 Implementing Architect Intent and Forward-Feed AIC Interaction Framework for addressing train derailment caused by adversarial drones. Table H.19 Mapping AIC interactions of the Eagle Drone and adversarial drone behaviours in mitigating train derailment risks.

	H.7 Stage 3B: Comprehensive Operational Environment Definition. Table H.35 Operational Design Definition for Eagle Robot Deployment in Train Track Zone.
	Validation is done by documenting expert reviews of architect assertions and predictions and the appropriate application of STCoT. A validation report template has been generated, which can be found No validation report had been generated as part of PhD scope.

L.4.1.2 Argument Pattern for AVOID System Operating Context Assurance

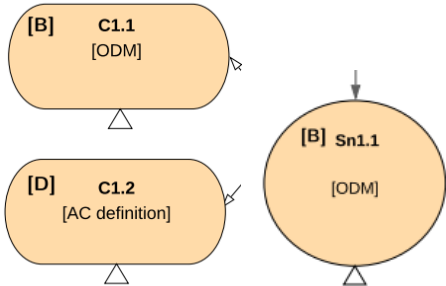
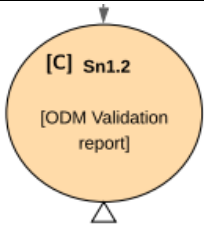
Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts; The following process outputs satisfy the artefact's demonstration requirements:
	Table I.13 Architect High-Level Solution Prescription. Table I.27 Operational Design Definition for AVP
	L.2 STCoT Validation Report for AVOIDDS case study. (I.4.1.1 Step 1.1) Identify a list of unsafe or confusing behaviours]
	I.4.4.3 Step 4.3) Define the assumptions and hazards made about factors.

	Table K.2 Extended assumptions, plausibility, concern and hazards analysis. In no particular order.
	I.4.3.1 Step 3.1) Model detailed AIC interactions scenarios for the problem domain. Figure I.2 Modelling AIC scenario from interaction n1.
	I.4.3.2 Step 3.2) Predict the extended list of emergent AIC interactions scenarios. Table I.10 AIC extended scenario for n1 Interaction STCoT definition. Table K.2 Extended assumptions, plausibility, concern and hazards analysis. In no particular order.
	I.6.1 Predictive Thinking Pipeline 1: Introducing Autonomous systems into Feed-forward complexity. Table I.14 Implementing Architect Intent and Forward-Feed AIC Interaction Framework for addressing AVP reliability for by-passing aircraft. Table I.15 Mapping AIC interactions of AVP with ownship aircraft and the environment
	I.7 Stage 3B: Comprehensive Operational Environment Definition. Table I.27 Operational Design Definition for AVP

	Validation is done by documenting expert reviews of architect assertions and predictions and the appropriate application of STCoT. A validation report template has been generated, which can be found No validation report had been generated as part of PhD scope.
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Table L.2 captures the primary artefacts that need to be presented:

Table L.2 SACE Stage 1 artefacts and AIC approach mapping

SACE Artefact	Explanation	The substantiating AIC methods and artefacts
[A]: AS Concept Definition	entails producing a high-level document outlining the intended functions, objectives, and constraints. The systems approach artefact should demonstrate that the intended system functionality, scope of autonomy, and interactions with humans are clearly defined. It should include a use-case description and stakeholder involvement agreement.	- Sections 6.3, M.3. - H,I.5 Stage 2: Architect Intent and Autonomous Solution Needs Definition method satisfies artefacts [A]. This is because the method for defining the architect's intent specifically delineates how the architect plans to address the anticipated problematic situations identified in Stage 1 and how autonomous functionalities may provide solutions for those issues. This stage encompasses establishing a contractual agreement between the architect and the stakeholder regarding the requirements to be fulfilled by the architect. For

		instance, the following may be observed: • Table H.15 - Architect High-Level Solution Prescription • Table I.13 - Architect High-Level Solution Prescription
[B]: Operational Domain Model (ODM)	entails modelling the scope of operations for AS, including the assumptions made about the environment and operating conditions. The systems approach artefact should demonstrate: • All relevant environmental and operational conditions are included. • All scenarios the AS may encounter are covered. • It should explicitly list assumptions, constraints, and non-mission interactions.	• Sections 6.2, M.2. • H,I.4 Stage 1: Uncertainty Problem Articulation and Operational Environment Modelling. In Stage 1, the architect methodically formulates a comprehensive Operational Domain Model (ODM) by clearly delineating uncertainties related to environmental conditions and contextual constraints that may be deemed critical to the safety of autonomous systems. This stage employs structured activities, such as the Predictive Thinking Pipeline (H.4.4), which assesses pivotal assumptions and factors related to the problem domain, ensuring a thorough operational environment representation. For instance, the Operational Design Definitions presented in Table H.35 (Eagle Robot

		Deployment in Train Track Zone) and Table I.27 (Automated Valet Parking, AVP) illustrate the sources of uncertainty the architect perceives. Furthermore, the complexity inherent in the operational domain is acknowledged. Consequently, Stage 1 asserts that the resultant ODM is adequately detailed and resilient, thereby helping minimise unforeseen occurrences upon the deployment of the autonomous system. It distinctly identifies the assumptions and variances necessary for subsequent safety analysis and assurance while explicitly addressing potential risks introduced by emergent complexity and operational nuances uncertainty.
[C]: ODM Validation Report	entails documenting the validation of plausibility and correctness of the ODM and that it sufficiently defines the operating scope. The architect must provide clear evidence that the ODM is comprehensive and correct. Since we utilise STCoT to model the ODM, validating it	Sections L.1 for STCoT validation process, The structured STCoT Validation Report Template (Artefact L.1), along with its specific instances about the AVOIDDS system (L.2) and Unsafe Train Tracks (L.3), systematically fulfils the SACE demonstration requirements.

	becomes more objective and straightforward. For such end, we developed a structured STCoT validation report template (section L.1).	Specifically, the AIC method distinctly delineates: • Validation objective of Completeness: The validation report examines whether the set of predictions is complete in implementing the thinking method and those general systems rules enforced; thus, the architect predicts all they can or need to predict. • Validation objective of Correctness: The validation report requires a formal review process to ensure the predictions regarding implementing the thinking method and general systems rules (axioms) are correct. • Validation objective of plausibility: The validation report examines whether STCoT predictions align with possible real-world complexity. Consequently, the structured AIC substantiating method meets the SACE artefact requirements by ensuring that the Operating Domain Model is thoroughly validated for
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		relevance, accuracy, completeness, and suitable granularity, with documented expert review outlined in the validation review process (L.1.6). The validation report template can be used to validate any AIC model schema.
[D]: Autonomous Capabilities Definition	entails specifying AS functionalities. The systems approach artefact is expected to clearly define the scope of autonomy and the tasks that may require human intervention.	• Architect High-Level Solution Prescription (Table H.15, I.13). • H,I.6 Stage 3A: HazTOPs and Ordered AIC-driven Autonomous System Requirements Development Stage 2 of the AIC approach develops the high-level solution for identified problematic situations and assumptions derived from stage 1. At this stage, the architect and stakeholders decide what autonomous capabilities need to be used. Stage 3A further refines the system level needs into more granular definitions of autonomous capabilities.
[E]: Operating Scenarios Definition	entail a detailed description of all scenarios the AS may encounter within the anticipated ODM. The systems	• H,I.6 Stage 3A: HazTOPs and Ordered AIC-driven Autonomous System

	approach artefact must define actions, events and environmental assumptions that could affect AS performance.	Requirements Development • H,I.7 Stage 3B: Comprehensive Operational Environment Definition. • H,I.8 Stage 4: Disordered AIC-Driven Black Swan Scenarios Prediction The AIC approach predicts scenarios at every stage of the design process and translates those scenarios into design requirements. One important activity is predicting Black Swan scenarios, which are part of the long-tailed probability distribution of possible events that could happen in the operational domain. The development stages define the actions, events and operational environment assumptions constituting the operational scenarios.
[F]: Operating Scenarios Validation Report	assesses whether the defined operating scenarios comprehensively capture all relevant AS interactions. The substantiating artefact must provide evidence of expert review, simulation-based verification, and real-world validation data to confirm the	• Sections L.1 for STCoT validation template, sections H.12, L.2 for implementations. Validation is done by documenting expert reviews of architect assertions and predictions and the appropriate application of

	completeness of the operating scenarios.	STCoT. A validation report template has been generated, which can be found. No validation report was generated as part of PhD scope.
[G]: AS Operating Context Assurance Argument Pattern & [H]: AS Operating Context Assurance Argument:	A structured assurance argument framework is explicitly formulated to demonstrate that the Autonomous System (AS) can safely operate within its specifically defined operational context.	The assurance argument framework presented in artefact L.4.1.1 (Argument Pattern for Eagle Robot Deployment) explicitly addresses the requirement by structuring the safety justification around systematically established operational domain assumptions and scenarios.

L.4.2 Stage 2: AS Hazardous Scenario Identification

SACE Stage 2 and AIC Stage 3A emphasise identifying and validating hazardous scenarios. The main outcomes of this stage:

1. Potential hazardous scenarios for the AS are identified and outlined.
2. Hazardous Scenarios of the AS are Validated.
3. Development of the Assurance Argument for the AS Hazardous Scenarios.

The AIC approach introduces HazTOPs (Hazards, Threats, and Opportunities Scenarios) to refine the scope of risk mitigation strategies.

AIC Systems Approach: Processes involved to satisfy the objectives:

- Stage 3A: HazTOPS and Ordered AIC-driven Autonomous System Requirements Development

- L.1 STCoT Validation Report Template

L.4.2.1 Argument Pattern for Eagle Robot Hazardous Scenarios

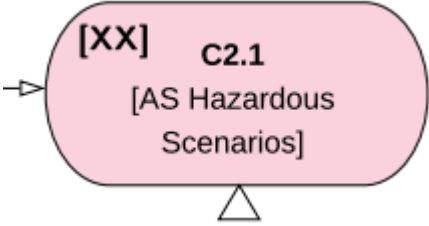
Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts; The following process outputs satisfy the artefact's demonstration requirements:
	H.6.3 Predictive Thinking Pipeline 3: Hazards, Threats and Opportunities Scenarios (HazTOPs) Analysis. Figure H.16 Hazards Complexity Field Scope: graphically scoping the hazards within the complexity field by placing hazard icons on target interaction. Figure H.17 Threats Complexity Field Scope Figure H.18 Opportunities Complexity Feels Scope H.6.3.2 Step 2) Characterise the scoped interactions. Figure H.19 Hazards associated with Eagle Drone preventing derailed train complexity field Table H.24 The table describes the AIC interaction dynamics between Eagle Drones and adversarial drones H.6.3.3 Step 3) Apply predictive potential complications guide words.

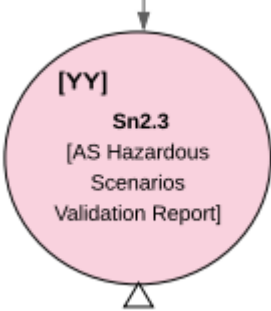
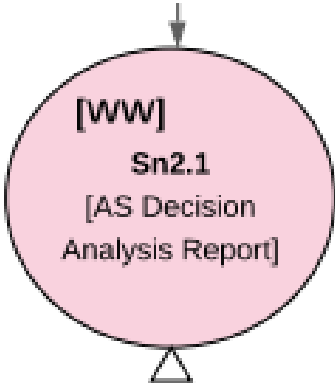
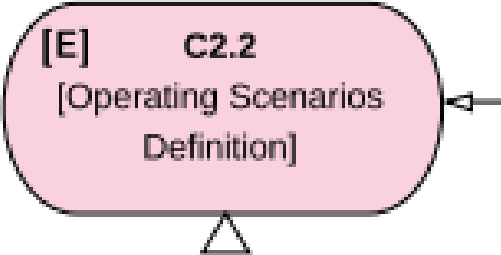
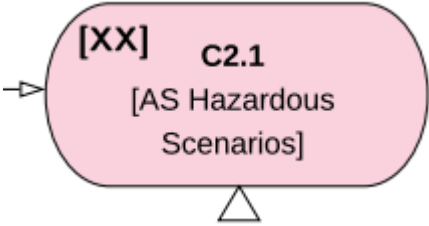
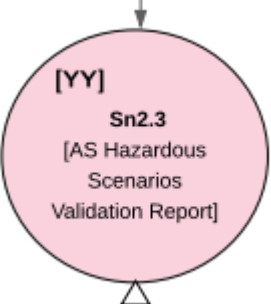
	Table H.25 HazTOPs Analysis of "3 Drones Attack" Scenario with Risk and Surprise Assessment Table H.26 HazTOPs Analysis of adversarial drone using smart lasers scenario Table H.27 HazTOPs Analysis of adversarial drone hiding behind fence scenario
	Validation is done by documenting expert reviews of architect assertions and predictions and the appropriate application of STCoT. A validation report template has been generated, which can be found in L.1 STCoT Validation Report Template Examples: L.3 STCoT Validation Report for Unsafe Train Tracks case study
	H.6.1 Predictive Thinking Pipeline 1: Introducing Autonomous systems into Feed-forward complexity H.6.2 Predictive Thinking Pipeline 2: Designing the affecting Backward-Feed complexity field H.9 Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Epistemic Intelligence Development
	H.6.1 Predictive Thinking Pipeline 1: Introducing Autonomous systems into Feed-forward complexity. Table H.18 Implementing Architect Intent and Forward-Feed AIC Interaction Framework for addressing train derailment caused by adversarial drones.

	Table H.19 Mapping AIC interactions of the Eagle Drone and adversarial drone behaviours in mitigating train derailment risks.
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L.4.2.2 Argument Pattern for AVOID System Hazardous Scenarios

Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts; The following process outputs satisfy the artefact's demonstration requirements:
	I.6.3 Predictive Thinking Pipeline 3: Hazards, Threats and Opportunities Scenarios (HazTOPs) Analysis. Figure I.11 Sources of Hazards AIC Complexity Field H.6.3.2 Step 2) Characterise the scoped interactions. Table I.22 Considering Hazards related to I1 interaction H.6.3.3 Step 3) Apply predictive potential complications guide words. Table I.23 Example “More” guide word complication
	Validation is done by documenting expert reviews of architect assertions and predictions and the appropriate application of STCoT. A validation report template has been generated, which can be found in L.1 STCoT Validation Report Template Examples:

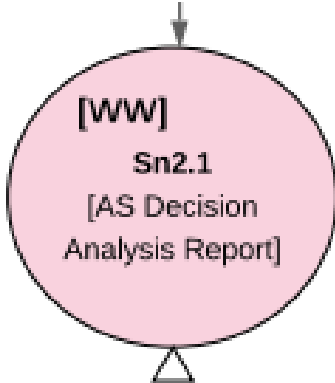
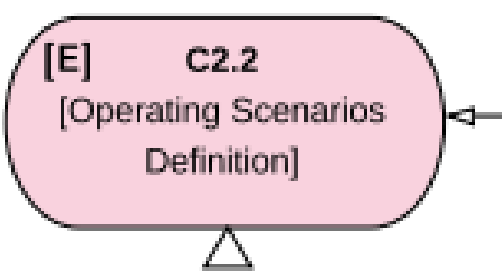
	L.2 STCoT Validation Report for AVOIDDS case study
	I.6.1 Predictive Thinking Pipeline 1: Introducing I.6.2 Predictive Thinking Pipeline 2: Designing the affecting Backward-Feed complexity field I.9 Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Epistemic Intelligence Development
	I.6 Stage 3A: HazTOPs and Ordered AIC-driven Autonomous System Requirements Development. Table I.14 Implementing Architect Intent and Forward-Feed AIC Interaction Framework for addressing AVP reliability for by-passing aircraft. Table H.19 Mapping AIC interactions of the Eagle Drone and adversarial drone behaviours in mitigating train derailment risks.

Table L.3 captures the primary artefacts that need to be presented in the safety case:

Table L.3 SACE Stage 2 artefacts and AIC approach mapping

SACE Artefact	Explanation	The substantiating AIC methods and artefacts
[WW]: AS Decision Analysis Report:	A report analysing the decisions made by the AS at key decision points within different operating scenarios.	I, H.6.1 Predictive Thinking Pipeline 1: Introducing Autonomous systems

	The systems approach artefact must demonstrate that all AS decisions and judgments have been identified and analysed for potential hazards and belief states of AS.	into Forward-Feed complexity • I,H.6.2 Predictive Thinking Pipeline 2: Designing the affecting Backward-Feed complexity field In this process, the architect determines an AS's decisions when involved in a particular problematic scenario. For example, • Table H.18 Implementing Architect Intent and Forward-Feed AIC Interaction Framework for addressing train derailment caused by adversarial drones The Eagle Drone should decide to inhibit the adversarial drone and prevent derailment of the train incident. However, to do so it must perform the following activities: • Recognises the visibility of vegetation appearance. • Avoids crashing into vegetation structures. • Physically inhibit the by-passing drone.
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		• It cannot effectively be influenced or controlled by the adversarial drone. Also, I,H.9 Stage 5: Safety-Driven ML-based Perception Training, Testing and Validation Process (CuneiForm Strategy Development) In this stage, the architect determines what decision AS should make when face a particular scenario, for example, in a situation such as (Table H.39 CuneiForm Pictorial situation articulation): A drone landing on the ground {1}. The drone has a camouflaged skin {2} landing at various distances from the track fence or train tracks or both {3}, trees {4}, local birds surrounding the landed drone{5} bushes {6}, gravel{7}, soil{8}, pavement{9} trash {10}. Then, the Eagle Drone should be able to recognise an adversarial drone and act accordingly. We do not specifically define a category of knowledge as "decision";
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		however, any action required and derived through the process from the ML component to perform is a decision made.
[XX]: AS Hazardous Scenarios Definition:	A comprehensive specification of all identified hazardous scenarios, including the interactions, environment states, and decisions leading to unsafe outcomes. The systems approach artefact should clearly show how the architect predicted hazardous scenarios and what mitigation requirements were devised to reduce their impact	• H.6 Stage 3A: HazTOPs and Ordered AIC-driven Autonomous System Requirements Development • H.8 Stage 4: Disordered AIC-Driven Black Swan Scenarios Prediction Stages 3A and 4 enable the architect to assess the problematic scenarios identified in Stage 1 thoroughly and discover additional scenarios. HazTops adopts a holistic approach that pinpoints hazards while identifying threats and opportunities. For instance, in Figure H.16, we modelled the operational scenario where a desired autonomous capability was defined, in which a hazard was predicted due to tree cover around the tracks, as it would obstruct the Eagle Drone's perception to identify. However, it can also be viewed as an opportunity because trees can complicate

		adversarial drone missions and can be utilised to provide an advantage for the Eagle Drone to ambush adversarial drones. Meanwhile, we identified a Soft Hazard concerning the relationship between agitated locals and the presence of surveillance robots, using the destruction of nature as a pretext to obstruct the continuation of the Eagle Drone's operations
[YY]: AS Hazardous Scenarios Validation Report:	This validation document confirms the completeness and correctness of the identified hazardous scenarios. The systems approach artefact should demonstrate that potential hazards are comprehensively covered and that all have been accounted for in the design.	The outputs of all predictive thinking pipelines and design steps provide comprehensive justification and design traceability for hazardous scenarios, including Black Swan scenarios. Validation is done by documenting expert reviews of architect assertions and predictions and the appropriate application of STCoT. A validation report template has been generated, which can be found in: • L.1 STCoT Validation Report Template • Examples: • L.2 STCoT Validation Report for AVOIDDS case study

		• L.3 STCoT Validation Report for Unsafe Train Tracks case study
[J]: AS Hazardous Scenarios Assurance Argument & [I]: AS Hazardous Scenarios Assurance Argument Pattern:	A structured assurance argument demonstrating that all hazardous scenarios have been sufficiently identified and validated.	• L.4.2.1 Argument Pattern for Eagle Robot Hazardous Scenarios. • L.4.2.2 Argument Pattern for AVOID System Hazardous Scenarios.

L.4.3 Stage 3: Safe Operating Concept Assurance

This phase involves actions aimed at defining and validating the safe operating concept for an AS. The following are the primary outcomes of this stage:

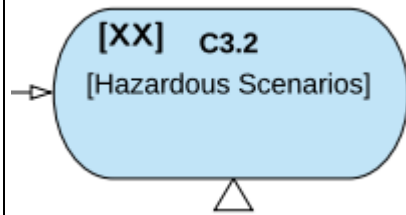
1. The Safe Operating Concept for the AS is clearly defined.
2. The Safe Operating Concept is validated.
3. Development of the Safe Operating Concept Assurance Argument.

AIC Systems Approach Processes involved to satisfy the objectives:

- Stage 3A: HazTOPS and Ordered AIC-driven Autonomous System Requirements Development
- Stage 3B: Comprehensive Operational Environment Definition

L.4.3.1 Argument Pattern for Eagle Robot SOC Assurance

Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts; The following process outputs satisfy the artefact's demonstration requirements:
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H.6.3 Predictive Thinking Pipeline 3: Hazards, Threats and Opportunities Scenarios (HazTOPs) Analysis.

Figure H.16 Hazards Complexity Field Scope: graphically scoping the hazards within the complexity field by placing hazard icons on target interaction.

Figure H.17 Threats Complexity Field Scope

Figure H.18 Opportunities Complexity Feels Scope

H.6.3.2 Step 2) Characterise the scoped interactions.

Figure H.19 Hazards associated with Eagle Drone preventing derailed train complexity field

Table H.24 The table describes the AIC interaction dynamics between Eagle Drones and adversarial drones

H.6.3.3 Step 3) Apply predictive potential complications guide words.

Table H.25 HazTOPs Analysis of "3 Drones Attack" Scenario with Risk and Surprise Assessment

Table H.26 HazTOPs Analysis of adversarial drone using smart lasers scenario

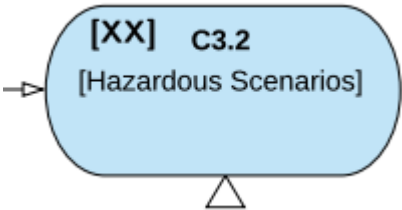
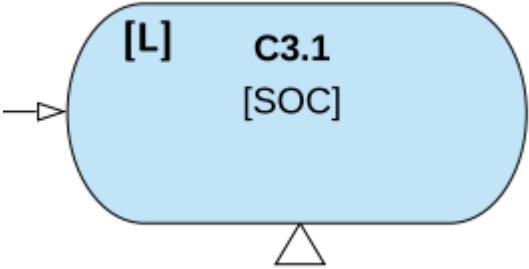
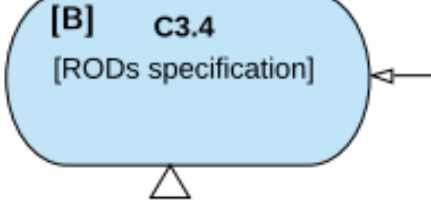
Table H.27 HazTOPs Analysis of adversarial drone hiding behind fence scenario

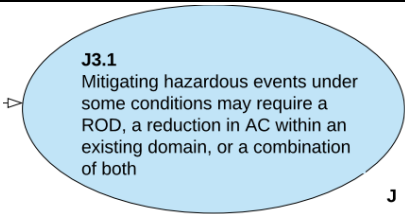
Figure H.20 Soft Hazard Complexity Field Model

A light blue rounded rectangle containing the text "[L] C3.1 [SOC]". It has an incoming arrow from the left and an outgoing arrow pointing downwards.	H.6.4.1,2 (c) Step 3) Ordered-AIC-based Mitigating System or Safety Requirements Derivation (Safety Concept) H.8.5 Step 5) Define mitigating ML Development and Safety Requirements.
A light blue circle containing the text "[M] Sn3.3 [SOC justification report]". It has an incoming arrow from the top and an outgoing arrow pointing downwards.	H.6.4 Predictive Thinking Pipeline 4:Elicitate AIC System-Level Requirements and Training Requirements
A light blue rounded rectangle containing the text "[B] C3.4 [RODs specification]". It has an incoming arrow from the right and an outgoing arrow pointing downwards.	To be completed [outside PhD scope]
A light blue oval containing the text "J3.1 Mitigating hazardous events under some conditions may require a ROD, a reduction in AC within an existing domain, or a combination of both". It has an incoming arrow from the left. A small 'J' is located at the bottom right of the oval.	To be completed [outside PhD scope]

L.4.3.2 Argument Pattern for AVOID System SOC Assurance

Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts; The following process outputs satisfy the artefact's demonstration requirements:
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	I.6.3 Predictive Thinking Pipeline 3: Hazards, Threats and Opportunities Scenarios (HazTOPs) Analysis. Figure I.11 Sources of Hazards AIC Complexity Field Figure H.17 Threats Complexity Field Scope Figure I.12 Sources of Threats AIC Complexity Field Table I.22 Considering Hazards related to I1 interaction Table I.23 Example “More” guide word complication
	I.6.4 Predictive Thinking Pipeline 4:Elicitate Ordered AIC System-Level Requirements and training requirements I.6.4.1,2 (c) Step 3) Ordered-AIC-based Mitigating System or Safety Requirements Derivation (Safety Concept)
	I.6.4 Predictive Thinking Pipeline 4:Elicitate Ordered AIC System-Level Requirements and training requirements
	To be completed [outside PhD scope]

	To be completed [outside PhD scope]
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The following are the primary artefacts that need to be presented:

Table L.4 SACE Stage 3 artefacts and AIC approach mapping

SACE Artefact	Explanation	The substantiating AIC methods and artefacts
[K]: Definition of Sufficiently Safe	Defines what constitutes an acceptable level of safety for the AS, considering legal, ethical, and stakeholder risk tolerance factors. The systems approach artefact must justify why the defined safety criteria are sufficient, referencing legal and regulatory guidelines, ethical considerations, and risk acceptance criteria.	The structure of STCoTs can be used as evidence to justify how the architect arrived at predicting hazards and Black Swan scenarios in the context of providing a comprehensive justification. However, justifying based on specific ethical standards or comparisons to human performance is outside the scope of the PhD. Nonetheless, STCoT is ethics-based on the application of universal systems rules about complexity. Ethical regulations can substitute those rules and thus provide systematic evidence of ethical design considerations for assurance purposes.
[L]: Safe Operating Concept Definition	A formal specification of how the AS must operate within its	The safety concept comprises a set of safety requirements at

	defined environment to ensure safety, incorporating necessary constraints and system safety requirements. The systems approach artefact must define specific system-level safety requirements, ensuring they are clear, unambiguous, and sufficient to mitigate hazardous scenarios.	the system level, not solely for the ML component. In our approach, we combine the system-level requirements with the training requirements (training concept), which do not precisely specify how to implement the training. S6age 4 provides further safety requirements based on mitigating Black Swan scenarios. The following is examples of how we captured safety concept and training concept: • H.6.4.1,2 (c) Step 3) Ordered-AIC-Based Mitigating System or Safety Requirements Derivation (Safety Concept) • H.6.4 Predictive Thinking Pipeline 4:Elicitate AIC System-Level Requirements and Training Requirements • H.8.5 Step 5) Define mitigating ML Development and Safety Requirements.
[M]: Safety Concept (SOC) Justification Report	A structured report validating that the SOC sufficiently mitigates the identified hazardous scenarios. The systems approach artefact	Stages 3A and 4 provide the comprehensive justification of how the architect arrives at their predictions and what

	must systematically justify how each safety requirement and operational constraint contributes to mitigating specific hazardous scenarios.	mitigation plans should be considered. For example, • I,H.6.3 Predictive Thinking Pipeline 3: Hazards, Threats and Opportunities Scenarios (HazTOPs) Analysis • I, H.6.4 Predictive Thinking Pipeline 4:Elicitate AIC System-Level Requirements and Training Requirements
[N]: SOC Assurance Argument Pattern & [O]: SOC Assurance Argument:	A structured framework to argue that the Safe Operating Concept sufficiently mitigates all hazardous scenarios identified in previous stages. The systems approach artefact must systematically justify how each safety requirement and operational constraint contributes to mitigating specific hazardous scenarios.	We constructed the GSN argument in the following sections: • L.4.3.1 Argument Pattern for Eagle Robot SOC Assurance. • L.4.3.2 Argument Pattern for AVOID System SOC Assurance

L.4.4 Stage 4: Safety Requirements Assurance

SACE stage 4 revolves around demonstrating that the safety requirements are comprehensively and correctly captured. The primary outcome of this stage is:

1. Safety requirements for each tier of the requirements development is clearly defined.
2. The defined safety requirements are validated.

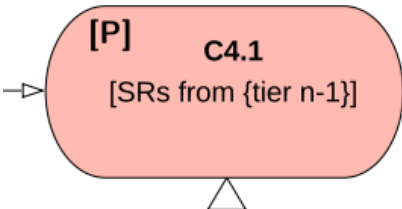
3. Development of Safety Requirements Argument.

AIC Systems Approach Processes involved to satisfy the objectives:

The pattern is mandated for each tier of the system approach. Therefore, the following structure would be:

- **Tier n-3:** Stage 2: Architect Intent and Autonomous Solution Needs Definition. Sections: I.5, H.5.
- **Tier n-2:** Stage 3A: HazTOPS and Ordered AIC-driven Autonomous System Requirements Development. Sections: I.6, H.6.
- **Tier n-1:** Stage 4: Disordered AIC-Driven Black Swan Scenarios Prediction. Sections: I.8, H.8.
- **Tier n:** Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Epistemic Intelligence Development. Sections: I.9, H.9.

L.4.4.1 Argument Pattern for Eagle Robot Safety Requirements Assurance

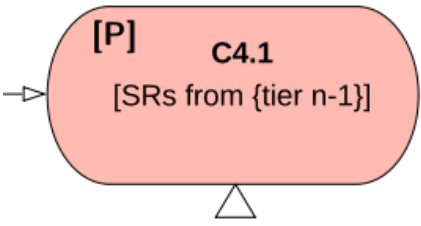
Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts; The following process outputs satisfy the artefact's demonstration requirements:
 Given: Tier n: Safety Driven CuneiForm Characterisation Then Tier n-1 : Safety ML Development Requirements (Training Concept)	Stage 3A, (c) Step 3) Ordered-AIC-based Mitigating System or Safety Requirements Derivation (Safety Concept). Table H.37 Eagle Drone Safety Training Requirements for Black Swan Scenarios Table H.38 ML component training dataset requirements

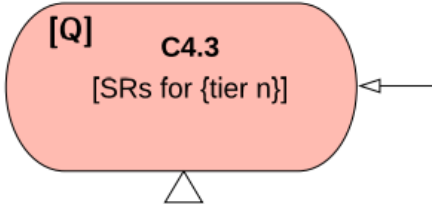
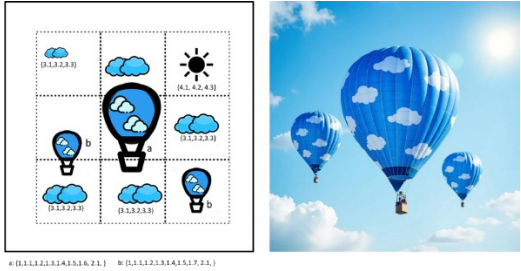
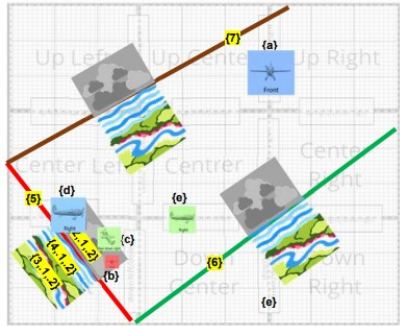
Tier n: Safety Driven CuneiForm Characterisation	Table H.39 CuneiForm Pictorial situation articulation Table H.40 Characteristic Training Classes definitions for a CuneiForm abstract image Figure H.36 Example output CuneiForm with appropriate instantiation using a simple CuneiForm canvas template example. We defined a Black Swan Scenarios Validation Dataset.
	Table H.37 Eagle Drone Safety Training Requirements for Black Swan Scenarios Figure H.36 Example output CuneiForm with appropriate instantiation using the CuneiForm canvas template example. We defined a Black Swan Scenarios Validation Dataset.

	Table H.35 Operational Design Definition for Eagle Robot Deployment in Train Track Zone Table H.31 4HnWs method for Eagle Drone adjusting patrol functionality Table H.15 Architect High-Level Solution Prescription Table H.16 Architect High-Level Solution Prescription related to the impact of roaming adversarial drones Table H.17 Architect High-Level Solution Prescription related to the police incapability to capture adversarial drone
	All stages, from stage 1 to stage 5,

L.4.4.2

Argument Pattern for AVOID Safety Requirements Assurance

Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts; The following process outputs satisfy the artefact's demonstration requirements:
	Table I.31a AVP Training Requirements for Black Swan Scenarios Table I.31b ML Safety Training Requirements and Perception Dataset Specifications for AVP

Tier n-1 : Safety ML Development Requirements (Training Concept)	
[Q] C4.3 [SRs for {tier n}]  Tier n: Safety Driven CuneiForm Characterisation	Table I.32 CuneiForm Pictorial situation articulation Table I.33 Characteristic Training Classes definitions for a CuneiForm abstract image Figure I.20 Example CuneiForm and instantiated image  Appendix D Safety Validation report for AVOIDDS dataset. Pages: 15, 19, 23, 27, 31, 35. For example, the following is safety-driven CuneiForm characterisation of a training dataset. 

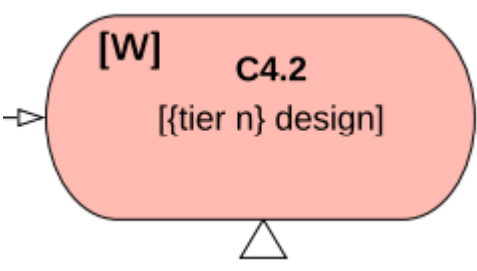
	Table I.13 Architect High-Level Solution Prescription Table I.26 Safety requirements derivations to mitigate the concealed drone problem. Table I.27 Operational Design Definition for AVP Table I.31a AVP Training Requirements for Black Swan Scenarios Figure I.20 Example CuneiForm and instantiated image
	All stages, from stage 1 to stage 5.

Table L.5 captures primary artefacts that need to be presented:

Table L.5 SACE Stage 4 artefacts and AIC approach mapping

SACE Artefact	Explanation	The substantiating AIC methods and artefacts
[P]: Safety Requirements from tier n-1 & [Q]: Safety Requirements for tier n:	The safety requirements are defined at the higher level tier of decomposition, which must be correctly allocated and interpreted at the current tier.	in the AIC systems approach, the tier-n-1 is relative; for example, the Stage 3A, (c) Step 3) Ordered-AIC-based Mitigating System or Safety

	The systems approach artefact must demonstrate that higher-level safety requirements are adequately decomposed, ensuring consistency and completeness in allocating to system components	Requirements Derivation (Safety Concept). Would be the tier n-1 for (d) Step 4) Extended Concrete Safety Concept and ML Safety Training Concept, output (tier n) Also, Given: Tier n: Safety Driven CuneiForm Characterisation Then, Tier n-1 : Safety ML Development Requirements (Training Concept) Therefore, the following would be evidence to be provided:
[R]: Safety Requirements Justification Report	A structured report validating that the decomposed safety requirements adequately maintain the intent of the original requirements. The systems approach artefact must provide traceability and justification for each safety requirement, ensuring they correctly address the identified hazards and are appropriately assigned to system components.	All stages, from stage 1 to stage 5, operate based on tight traceability. They also operate on the premise of exposing the architectural thought process that informs any decision, thus providing a very clear insight into the justification behind design choices and the principles used to make such engineering decisions. For example: H, I.6 Stage 3A: HazTOPs and Ordered AIC-driven

		Autonomous System Requirements Development Since we are including CuneiForms as the implementation of SOC, then the following tables will provide the justification required (examples): Table M.21 Characteristic Training Classes definitions for a CuneiForm abstract image (AVOIDDS) Table M.20 CuneiForm Pictorial situation articulation (AVOIDDS) Table M.19 ML Safety Training Requirements and Perception Dataset Specifications for AVP Table M.18 AVP Training Requirements for Black Swan Scenarios (AVOIDDS)
[W]: tier n Design:	The design specification at the current tier of decomposition defines system components and their interactions. The systems approach artefact must explicitly define design requirements which meet the higher-level safety requirements.	These are the more concrete requirements or specifications directly relating to higher-level safety requirements. For example, Table H.31 4HnWs method for Eagle Drone adjusting patrol functionality.

		• Figure I.20 Example CuneiForm and instantiated image. We would also include the following tables as evidence: Table 7.10 Black Swan Scenarios Batch A and B CuneiForms Table 7.11 Typical operations CuneiForms Figure 7.6 H.54 Out-of-context CuneiForm of drones. Figure M.20 Example CuneiForm and instantiated image for AVOIDDS case study They represent the concrete level implementation of safety concept at dataset level.
[S]: Safety Requirements Argument Pattern & [T]: Safety Requirements Argument:	A structured framework for demonstrating that the safety requirements at each tier adequately capture the intent of the previous tier's requirements.	We captured the patterns for this stage in the following sections: • L.4.5.1 Argument Pattern for Eagle Robot Design Assurance • L.4.5.2 Argument Pattern for AVOID system Design Assurance

L.4.5 Stage 5: AS Design Assurance

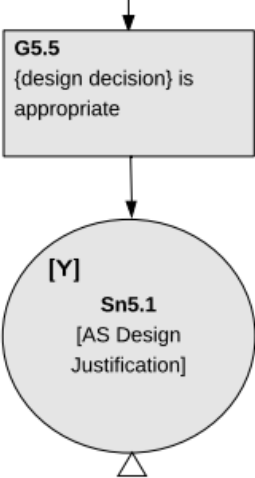
This stage focuses on assuring the concrete design. It is iterative in nature, accounting for the AS's design assurance at various design decomposition levels. The following are the main outcomes:

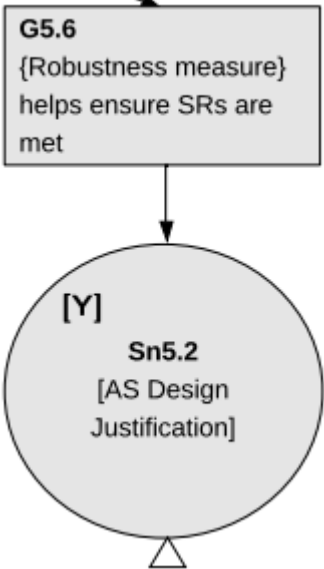
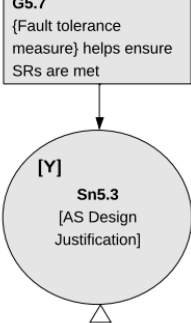
1. Design at tier n, which satisfies safety requirements, is specified.
2. The sufficiency of the design at tier n is justified.
3. The claim of sufficiency is validated.
4. Development of As design assurance argument.

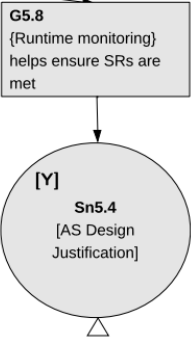
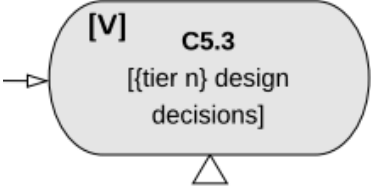
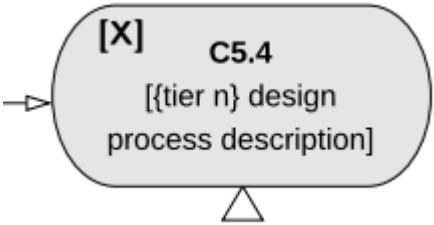
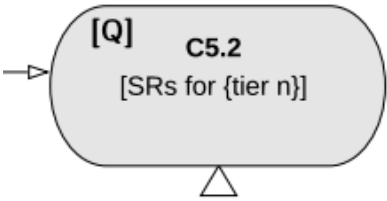
AIC Systems Approach Processes involved to satisfy the objectives:

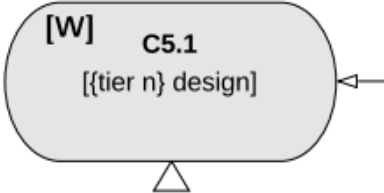
- All AIC Systems Approach stages.

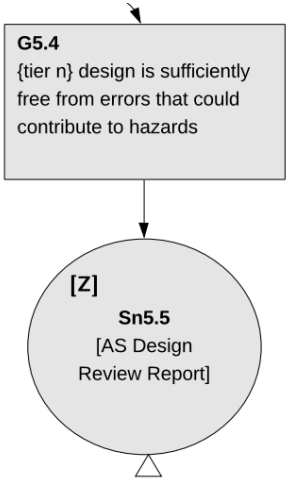
L.4.5.1 Argument Pattern for Eagle Robot Design Assurance

Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts; The following process outputs satisfy the artefact's demonstration requirements:
 The diagram shows a goal node 'G5.5 {design decision} is appropriate' with an incoming arrow from above. An arrow points from this goal to a strategy node 'Sn5.1 [AS Design Justification]' which is enclosed in a circle labeled '[Y]'. The strategy node has an outgoing arrow pointing to a small triangle at the bottom. Justification that key design decisions taken at tier n are appropriate to meet the defined safety requirements.	All Predictive Thinking Pipelines and design steps' outputs

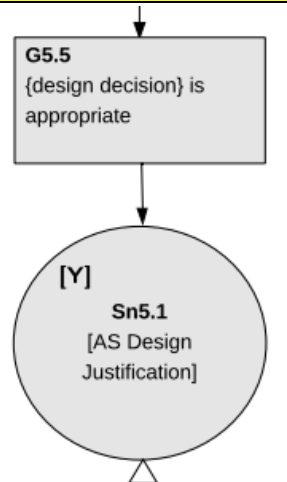
 G5.6 {Robustness measure} helps ensure SRs are met [Y] Sn5.2 [AS Design Justification] Demonstrating that the system has been designed to be robust against potential unexpected environmental changes.	Table H.37 Eagle Drone Safety Training Requirements for Black Swan Scenarios Figure H.36 Example output CuneiForm with appropriate instantiation using a simple CuneiForm canvas template example. We defined a Black Swan Scenarios Validation Dataset. Table H. 42 Comparison of Black Swan Lacked vs. Black Swan Enhanced Object Detection Experiments H.10.5 Experiment 1: Limited Live video-based experimentation H.10.6 Experiment 2: ML development environment-based validation
 G5.7 {Fault tolerance measure} helps ensure SRs are met [Y] Sn5.3 [AS Design Justification] Justification of fault tolerance mechanisms ensuring continued operation despite failures.	Outside PhD scope

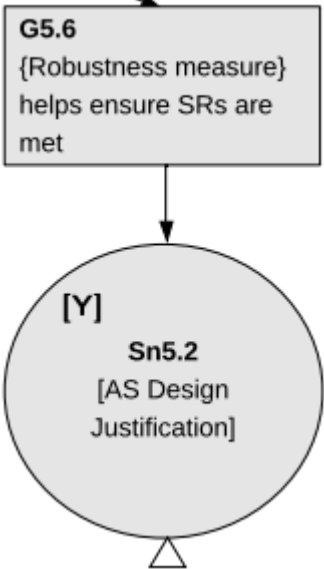
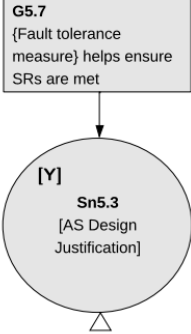
 Demonstrating that the system has an active monitoring mechanism to track and adapt its behavior based on operational feedback.	Outside PhD scope
 Presenting an argument over the appropriateness of each design decision to ensure system safety.	All STCoTs present a clear argument on the thought processes that went into
 Ensuring that the design process itself includes sufficient validation steps to prevent hazardous failures.	L.1 STCoT Validation Report Template L.3 STCoT Validation Report for Unsafe Train Tracks case study Stage (1-6) in AIC Systems Approach H.10.5 Experiment 1: Limited Live video-based experimentation H.10.6 Experiment 2: ML development environment-based validation
	Table H.33 Safety requirements derivations to mitigate the concealed drone problem. Table H.34 ML Safety Requirements Derivation

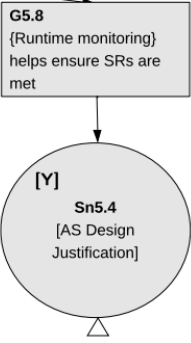
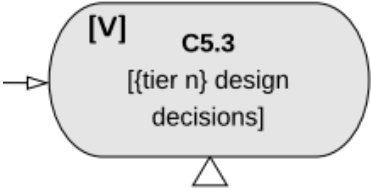
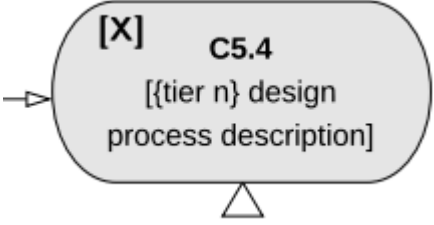
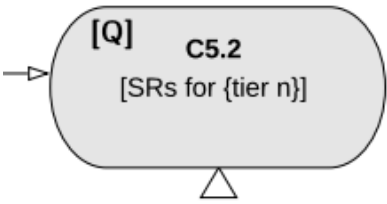
The context must demonstrate that the defined safety requirements at tier n comprehensively address operational risks, failure modes, and system constraints, ensuring the AS design meets safety assurance objectives and regulatory compliance.	Table H.37 Eagle Drone Safety Training Requirements for Black Swan Scenarios Table H.38 ML component training dataset requirements Figure H.36 Example output CuneiForm with appropriate instantiation using a simple CuneiForm canvas template example. We defined a Black Swan Scenarios Validation Dataset. 
	Table H.37 Eagle Drone Safety Training Requirements for Black Swan Scenarios Figure H.36 Example output CuneiForm with appropriate instantiation using the CuneiForm canvas template example. We defined a Black Swan Scenarios Validation Dataset. Table H.35 Operational Design Definition for Eagle Robot Deployment in Train Track Zone Table H.31 4HnWs method for Eagle Drone adjusting patrol functionality Table H.15 Architect High-Level Solution Prescription Table H.16 Architect High-Level Solution Prescription related to the impact of roaming adversarial drones Table H.17 Architect High-Level Solution Prescription related to the police incapability to capture adversarial drone

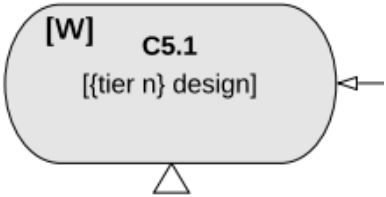
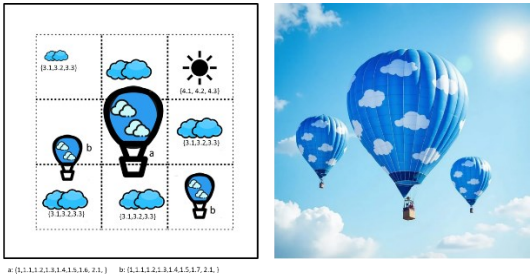
	Validation is done by documenting expert reviews of architect assertions and predictions and the appropriate application of STCoT. A validation report template has been generated, which can be found in L.1 STCoT Validation Report Template L.3 STCoT Validation Report for Unsafe Train Tracks case study
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L.4.5.2 Argument Pattern for AVOID System Design Assurance

Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts; The following process outputs satisfy the artefact's demonstration requirements:
 Justification that key design decisions taken at tier n are appropriate to meet the defined safety requirements.	All Predictive Thinking Pipelines and design steps' outputs

 Demonstrating that the system has been designed to be robust against potential unexpected environmental changes.	Table I.31a AVP Training Requirements for Black Swan Scenarios Table I.31b ML Safety Training Requirements and Perception Dataset Specifications for AVP I.9 Stage 5: Safety-Driven ML-based Perception Training, Testing and Validation Design (CuneiForm Strategy Development) I.10 Stage 6: Black Swan-driven ML Development and Testing
 Justification of fault tolerance mechanisms ensuring continued operation despite failures.	Outside PhD scope

 Demonstrating that the system has an active monitoring mechanism to track and adapt its behavior based on operational feedback.	Outside PhD scope
 Presenting an argument over the appropriateness of each design decision to ensure system safety.	All STCoTs present a clear argument on the thought processes that went into
 Ensuring that the design process itself includes sufficient validation steps to prevent hazardous failures.	L.1 STCoT Validation Report Template L.2 STCoT Validation Report for AVOIDDS case study Stage n in AIC Systems Approach
 The context must demonstrate that the defined safety requirements at	Given Tier n is the dataset: Table I.26 Safety requirements derivations to mitigate the concealed drone problem. Table I.31a AVP Training Requirements for Black Swan Scenarios

tier n comprehensively address operational risks, failure modes, and system constraints, ensuring the AS design meets safety assurance objectives and regulatory compliance.	Table I.31b ML Safety Training Requirements and Perception Dataset Specifications for AVP Figure I.18 We used DALL-E to generate this Black Swan Scenario for validation. Figure I.20 Example CuneiForm and instantiated image
	Table I.13 Architect High-Level Solution Prescription Table I.26 Safety requirements derivations to mitigate the concealed drone problem. Stage 3A, Step 4) Extended Concrete Safety Concept and ML Safety Training Concept Table I.27 Operational Design Definition for AVP Table I.31a AVP Training Requirements for Black Swan Scenarios Table I.31b ML Safety Training Requirements and Perception Dataset Specifications for AVP Figure I.20 Example CuneiForm and instantiated image 

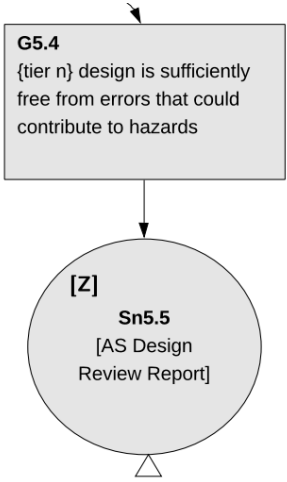
 G5.4 {tier n} design is sufficiently free from errors that could contribute to hazards [Z] Sn5.5 [AS Design Review Report]	Validation is done by documenting expert reviews of architect assertions and predictions and the appropriate application of STCoT. A validation report template has been generated, which can be found in L.1 STCoT Validation Report Template L.2 STCoT Validation Report for AVOIDDS case study
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Table 8.5 captures primary artefacts that need to be presented:

Table L.6 SACE Stage 5 artefacts and AIC approach mapping

SACE Artefact	Explanation	The substantiating AIC methods and artefacts
[V]: AS Development Log, [X]: Design Process for tier n, [Z]: AS Design Review	A comprehensive record of the design evolution, decisions, and iterations taken throughout development. The systems approach artefact must track all design changes, safety considerations, and iterations, ensuring full traceability of design decisions.	All STCoTs present a clear argument on the thought processes that went into all engineering judgements. This can be organised similarly as Appendix H and I. Also, we can add in the validation reporting that captures the design reviews outcomes.
[AA]: AS Design Assurance Argument & [U]: AS Design Assurance Argument Pattern:	we captured the GSN patterns in: • L.4.5.1 Argument Pattern for Eagle Robot Design Assurance • L.4.5.2 Argument Pattern for AVOID system Design Assurance.	

L.4.6 Stage 6: Hazardous Failures Management

SACE Stage 6 provides evidence demonstrating that the systems approach had thoroughly considered the hazards analysis. The primary outcomes of stage 6 are:

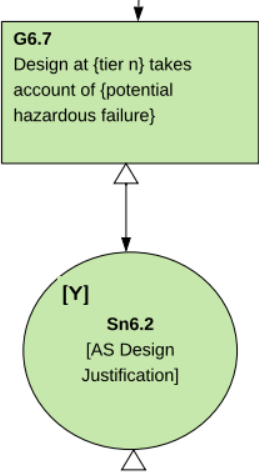
1. AS Hazardous Failures are identified.
2. The identified AS Hazardous Failures are mitigated.
3. Developing Hazardous Failures Assurance Argument Pattern

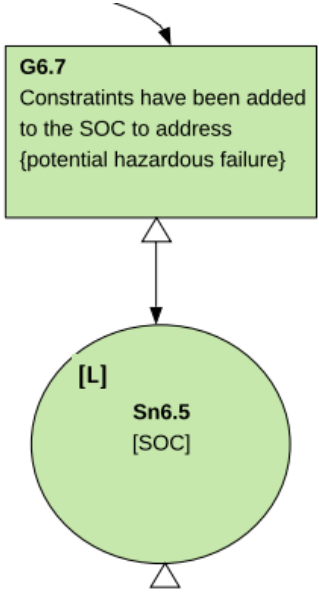
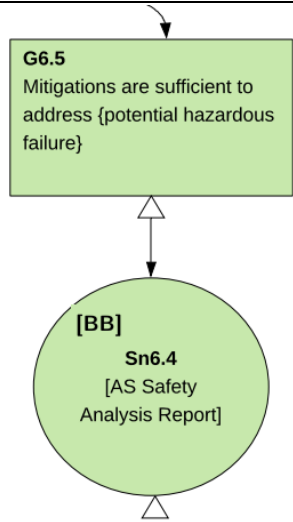
AIC Systems Approach Processes involved to satisfy the objectives:

- Stage 3A: HazTOPS and Ordered AIC-driven Autonomous System Requirements Development.
- Stage 3B: Comprehensive Operational Environment Definition.
- Stage 4: Disordered AIC-Driven Black Swan Scenarios Prediction.
- Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Epistemic Intelligence Development.

L.4.6.1 Argument Pattern for Eagle Robot Hazardous Failures

Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts; The following process outputs satisfy the artefact's demonstration requirements:
G6.6 Derived safety requirements have been derived to mitigate {potential hazardous failure} [Q] Sn6.3 [SRs for {tier n}]	Table H.33 Safety requirements derivations to mitigate the concealed drone problem. Table H.34 ML Safety Requirements Derivation Table H.37 Eagle Drone Safety Training Requirements for Black Swan Scenarios Table H.38 ML component training dataset requirements Figure H.36 Example output CuneiForm with appropriate instantiation using a simple CuneiForm

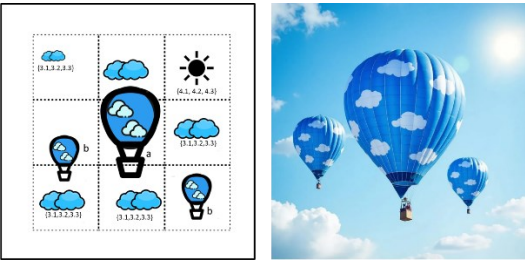
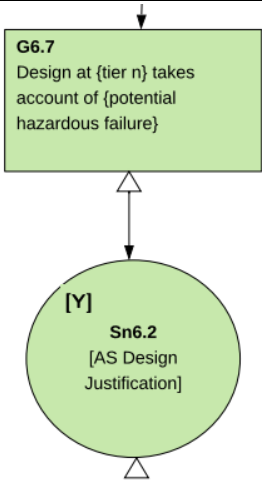
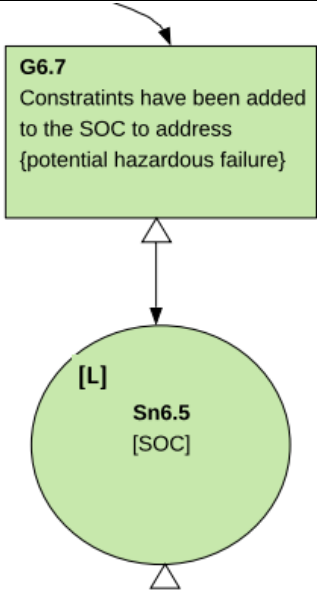
	canvas template example. We defined a Black Swan Scenarios Validation Dataset. 
	Table H.33 Safety requirements derivations to mitigate the concealed drone problem. Table H.34 ML Safety Requirements Derivation Table H.37 Eagle Drone Safety Training Requirements for Black Swan Scenarios Table H.39 CuneiForm Pictorial situation articulation Table H.40 Characteristic Training Classes definitions for a CuneiForm abstract image Figure H.36 Example output CuneiForm with appropriate instantiation using the CuneiForm canvas template example. We defined a Black Swan Scenarios Validation Dataset. 

 Within the context of maitaining Reduced Operating Domain (ROD)	We did not consider ROD in our approach for nw, hence it is Outside PhD scope
	H.6 Stage 3A: HazTOPs and Ordered AIC-driven Autonomous System Requirements Development H.7 Stage 3B: Comprehensive Operational Environment Definition H.8 Stage 4: Disordered AIC-Driven Black Swan Scenarios Prediction



L.4.6.2 Argument Pattern for AVOID system Hazardous Failures

Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts; The following process outputs satisfy the artefact's demonstration requirements:
	Given Tier n is the dataset: Table I.26 Safety requirements derivations to mitigate the concealed drone problem. Table I.31a AVP Training Requirements for Black Swan Scenarios Table I.31b ML Safety Training Requirements and Perception Dataset Specifications for AVP Figure I.18 We used DALL-E to generate this Black Swan Scenario for validation. Figure I.20 Example CuneiForm and instantiated image

	
	Table I.31a AVP Training Requirements for Black Swan Scenarios Table I.31b ML Safety Training Requirements and Perception Dataset Specifications for AVP I.9 Stage 5: Safety-Driven ML-based Perception Training, Testing and Validation Design (CuneiForm Strategy Development) I.10 Stage 6: Black Swan-driven ML Development and Testing
 Within the context of maitaining Reduced Operating Domain (ROD)	Outside PhD scope

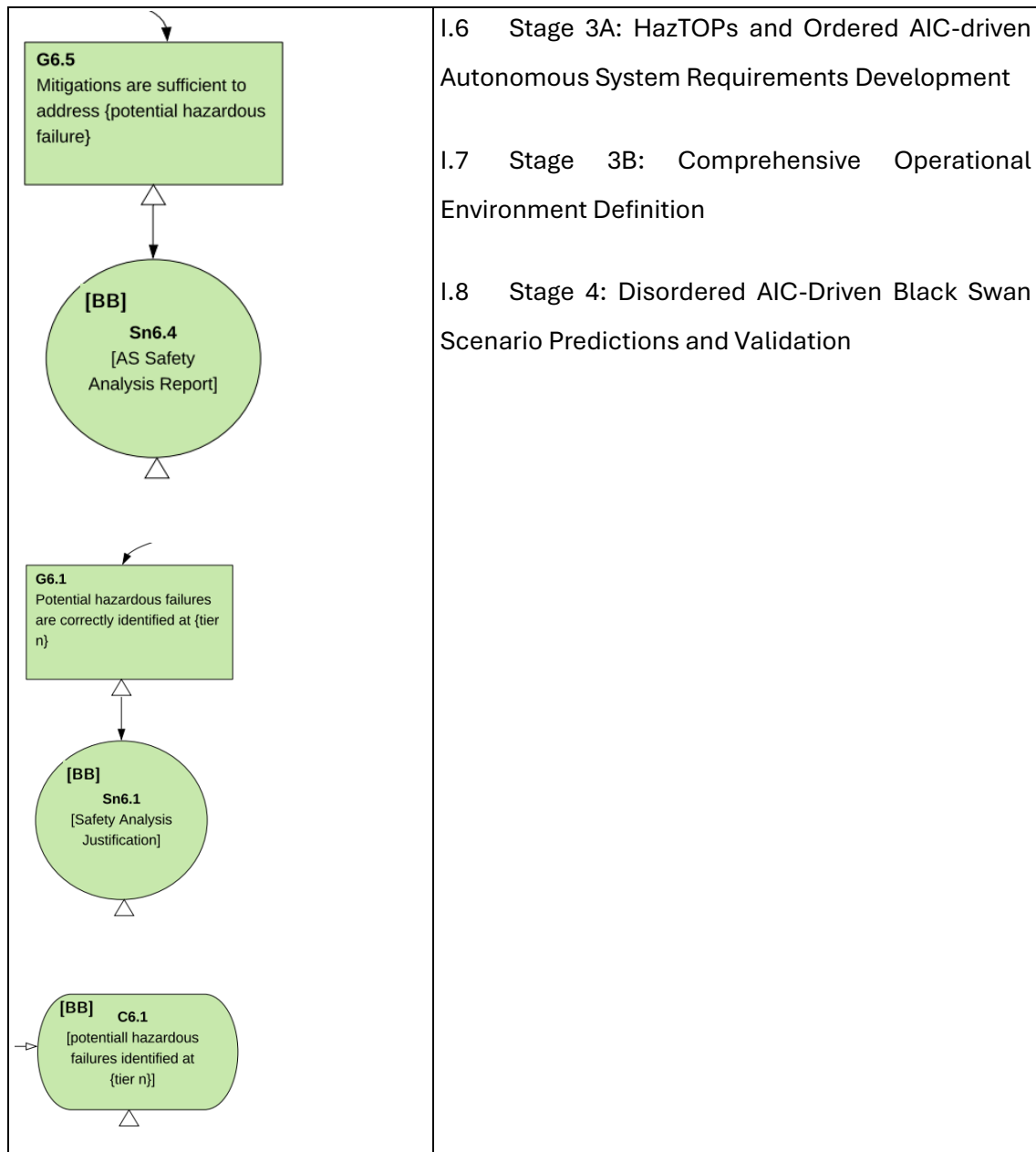


Table 8.6 captures primary artefacts that need to be presented:

Table L.7 SACE Stage 6 artefacts and AIC approach mapping

SACE Artefact	Explanation	The substantiating AIC methods and artefacts
[BB]: AS Safety Analysis Report:	A comprehensive report detailing the identified hazardous failures and the justification for the analysis approach used. The systems approach must document all	AIC systems approach includes the following techniques dedicated to discovering Hard and Soft hazards and then translating

	identified hazardous failures, the rationale for their identification, failure modes, potential consequences, and their impact on AS safety.	them into safety concepts and training concepts: • H.6 Stage 3A: HazTOPs and Ordered AIC-driven Autonomous System Requirements Development • H.7 Stage 3B: Comprehensive Operational Environment Definition • H.8 Stage 4: Disordered AIC-Driven Black Swan Scenario Predictions and Validation
[DD]: Hazardous Failures Argument Pattern & [EE]: Hazardous Failures Argument:	We captured the AIC outputs in the following sections: • L.4.6.1 Argument Pattern for Eagle Robot Hazardous Failures • L.4.6.2 Argument Pattern for AVOID System Hazardous Failures	

L.4.7 Stage 7: Out-of-Context Operation Assurance

Outside PhD scope

L.4.8 Stage 8: AS Verification Assurance

Outside PhD scope

L.4.9 Summary of SACE artefacts and AIC Systems Approach implementation

SACE Artefact	SACE Stage	Definition	General Demonstration Requirements	Substantiating AIC methods/Example: The following process outputs satisfy the artefact's demonstration requirements:
[A]: AS Concept Definition	1	High-level document outlining the system's intended functions, objectives, and constraints.	The systems approach artefact must define intended system functionality, scope of autonomy, and interaction with humans. Should include use-case descriptions and stakeholder agreements.	H,I.5 Stage 2: Architect Intent and Autonomous Solution Needs Definition Table H.15 Architect High-Level Solution Prescription Table I.13 Architect High-Level Solution Prescription
[AA]: AS Design Assurance Argument [U] : AS Design Assurance Argument Pattern	5	A structured assurance case demonstrates that tier n's design sufficiently satisfies the defined safety requirements.	The systems approach artefact must logically argue that design decisions at each tier ensure safety, referencing artefacts [Y], [Z], and [V] to provide supporting evidence.	L.4.5.1 Argument Pattern for Eagle Robot Design Assurance L.4.5.2 Argument Pattern for AVOID system Design Assurance
[B] : Operational Domain Model (ODM)	1	A model defining the scope of operation for the AS, including assumptions about the environment and	The systems approach artefact must include all relevant environmental and operational conditions, ensuring that all scenarios the AS may encounter are covered.	H,I.4 Stage 1: Uncertainty Problem Articulation and Operational Environment Modelling

		operating conditions.	It should explicitly list assumptions, constraints, and non-mission interactions.	Table H.35 Operational Design Definition for Eagle Robot Deployment in Train Track Zone Table I.27 Operational Design Definition for AVP
[BB] : AS Safety Analysis Report	6	A comprehensive report detailing the identified hazardous failures and the justification for the analysis approach used.	The systems approach artefact must document all identified hazardous failures, the rationale for their identification, failure modes, potential consequences, and their impact on AS safety.	H.6 Stage 3A: HazTOPs and Ordered AIC-driven Autonomous System Requirements Development H.7 Stage 3B: Comprehensive Operational Environment Definition H.8 Stage 4: Disordered AIC-Driven Black Swan Scenarios Prediction I.6 Stage 3A: HazTOPs and Ordered AIC-driven Autonomous System Requirements Development I.7 Stage 3B: Comprehensive Operational Environment Definition I.8 Stage 4: Disordered AIC-Driven Black Swan

				Scenario Predictions and Validation
[C] : ODM Validation Report	1	A documented validation of the completeness and correctness of the ODM, ensuring it sufficiently defines the operating scope.	Must provide evidence of review, simulation testing, and field validation to verify that all necessary operational elements have been captured. It should also justify the granularity level of the ODM.	L.1 STCoT Validation Report Template L.2 STCoT Validation Report for AVOIDDS case study L.3 STCoT Validation Report for Unsafe Train Tracks case study
[D] : Autonomous Capabilities Definition	1	A document specifying the AS's autonomous functionalities, limitations, and human-AS interaction boundaries.	The systems approach artefact must clearly define the scope of autonomy, specifying tasks the AS can perform independently and those requiring human intervention. Should also outline conditions under which autonomy is constrained.	Table H.15 Architect High-Level Solution Prescription. Table I.13 Architect High-Level Solution Prescription H,I.6 Stage 3A: HazTOPs and Ordered AIC-driven Autonomous System Requirements Development
[DD] : Hazardous Failures Argument Pattern	6	A structured framework for reasoning about identifying and mitigating	The systems approach artefact must demonstrate that hazardous failures have been	L.4.6.1 Argument Pattern for Eagle Robot Hazardous Failures

[EE]: Hazardous Failures Argument		hazardous failures at each tier of AS design.	systematically identified and addressed, ensuring that the AS design mitigates safety risks effectively.	L.4.6.2 Argument Pattern for AVOID system Hazardous Failures
[E] : Operating Scenarios Definition	1	A detailed description of all scenarios the AS may encounter within the defined ODM, including static and dynamic interactions.	The systems approach artefact must define actions, events, and environmental conditions affecting the AS. Should specify scenes (static conditions) and scenarios (temporal developments over time).	H.4.4.3 Step 4.3) Define the assumptions made about factors. Table J.8 unsafe train tracks problem domain assumptions. H.4.3.1 Step 3.1) Model detailed AIC interactions scenarios for the problem domain. Figure H.3 Modelling a complicated interaction n6. Table J.6 AIC problem domain scenarios definition H.4.3.2 Step 3.2) Predict the extended list of emergent AIC interactions scenarios. Table H.10 Complexity Field for n6 Interaction STCoT definition. Table J.7 AIC Extended Scenarios. H.6.1 Predictive Thinking Pipeline 1: Introducing Autonomous systems into Feed-forward complexity.

				Table H.18 Implementing Architect Intent and Forward-Feed AIC Interaction Framework for addressing train derailment caused by adversarial drones. Table H.19 Mapping AIC interactions of the Eagle Drone and adversarial drone behaviours in mitigating train derailment risks. H.7 Stage 3B: Comprehensive Operational Environment Definition. Table H.35 Operational Design Definition for Eagle Robot Deployment in Train Track Zone. I.4.4.3 Step 4.3) Define the assumptions and hazards made about factors. Table K.2 Extended assumptions, plausibility, concern and hazards analysis. In no particular order. I.4.3.1 Step 3.1) Model detailed AIC interactions scenarios for the problem domain.
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				Figure I.2 Modelling AIC scenario from interaction n1. I.4.3.2 Step 3.2) Predict the extended list of emergent AIC interactions scenarios. Table I.10 AIC extended scenario for n1 Interaction STCoT definition. Table K.2 Extended assumptions, plausibility, concern and hazards analysis. In no particular order. I.6.1 Predictive Thinking Pipeline 1: Introducing Autonomous systems into Feed-forward complexity. Table I.14 Implementing Architect Intent and Forward-Feed AIC Interaction Framework for addressing AVP reliability for by-passing aircraft. Table I.15 Mapping AIC interactions of AVP with ownship aircraft and the environment I.7 Stage 3B: Comprehensive Operational Environment Definition. Table I.27 Operational Design Definition for AVP
[F] : Operating Scenarios	1	A validation document	The systems approach artefact must provide	

Validation Report		assessing whether the defined operating scenarios comprehensively capture all relevant AS interactions.	evidence of expert review, simulation-based verification, and real-world validation data to confirm the completeness of the operating scenarios.	Validation is done by documenting expert reviews of architect assertions and predictions and the appropriate application of STCoT. A validation report template has been generated, which can be found No validation report had been generated as part of PhD scope.
[G] : AS Operating Context Assurance Argument Pattern [H] : AS Operating Context Assurance Argument	1	A structured assurance argument framework for ensuring the AS operates safely within its defined context.	The systems approach artefact must logically argue that the defined Operational Domain Model (ODM) and operating scenarios are sufficient, comprehensive, and correctly validated to support safe operations.	L.4.1.1 Argument Pattern for Eagle Robot Operating Context Assurance L.4.1.1 Argument Pattern for AVOID System Operating Context Assurance
[I] : AS Hazardous Scenarios Assurance Argument Pattern [J] : AS Hazardous	2	A structured framework ensures that all hazardous scenarios have been correctly identified and analysed.	The systems approach artefact must logically argue and provide evidence that all hazardous scenarios have been identified, relevant decisions analysed, and interactions between	L.4.2.1 Argument Pattern for Eagle Robot Hazardous Scenarios L.4.2.2 Argument Pattern for AVOID System Hazardous Scenarios

Scenarios Assurance Argument			the AS and its environment adequately considered.	
[K] : Definition of Sufficiently Safe	3	Defines what constitutes an acceptable level of safety for the AS, considering legal, ethical, and stakeholder risk tolerance factors.	The systems approach artefact must justify why the defined safety criteria are sufficient, referencing legal and regulatory guidelines, ethical considerations, and risk acceptance criteria. It may include comparisons to human operators or specific scenario-based safety assessments.	Outside PhD scope
[L] : Safe Operating Concept Definition	3	A formal specification of how the AS must operate within its defined environment to ensure safety, incorporating necessary constraints and system safety requirements.	The systems approach artefact must define specific system-level safety requirements, ensuring they are clear, unambiguous, and sufficient to mitigate hazardous scenarios. Should reference the Operational Domain Model (ODM) and the AS's autonomous capabilities.	H.6.4.1,2 (c) Step 3) Ordered-AIC-based Mitigating System or Safety Requirements Derivation (Safety Concept) H.6.4 Predictive Thinking Pipeline 4:Elicitate AIC System-Level Requirements and Training Requirements
[M] : SOC Justification Report	3	A structured report validating that the SOC sufficiently mitigates the identified	The systems approach artefact must systematically justify how each safety requirement and operational constraint	I,H.6.3 Predictive Thinking Pipeline 3: Hazards, Threats and Opportunities Scenarios (HazTOPs) Analysis

		hazardous scenarios.	contributes to mitigating specific hazardous scenarios. Stakeholder reviews, scenario simulations, and expert evaluations should be included as validation methods.	I, H.6.4 Predictive Thinking Pipeline 4:Elicitate AIC System-Level Requirements and Training Requirements
[N] : SOC Assurance Argument Pattern [O] : SOC Assurance Argument	3	A structured framework to argue that the Safe Operating Concept sufficiently mitigates all hazardous scenarios identified in previous stages.	The systems approach artefact must logically demonstrate that the SOC addresses all identified hazardous scenarios, ensuring that system safety requirements, reduced operating domains (RODs), and constraints sufficiently mitigate risks.	L.4.3.1 Argument Pattern for Eagle Robot SOC Assurance L.4.3.2 Argument Pattern for AVOID System SOC Assurance
[P] : Safety Requirements from tier n-1	4	The safety requirements are defined at the previous tier of decomposition, which must be correctly allocated and interpreted at the current tier.	The systems approach artefact must demonstrate that higher-level safety requirements are adequately decomposed, ensuring consistency and completeness in allocating system components.	(c) Step 3) Ordered-AIC-based Mitigating System or Safety Requirements Derivation (Safety Concept) Table H.37 Eagle Drone Safety Training Requirements for Black Swan Scenarios Table H.38 ML component training dataset requirements Table I.31a AVP Training Requirements for Black Swan Scenarios

				Table I.31b ML Safety Training Requirements and Perception Dataset Specifications for AVP
[Q] : Safety Requirements for tier n	4	The newly defined safety requirements at the current tier, allocated to relevant system components.	The systems approach artefact must prove that these safety requirements align with the previous tier's intent, effectively mitigating risks and guiding system design.	Table H.39 CuneiForm Pictorial situation articulation Table H.40 Characteristic Training Classes definitions for a CuneiForm abstract image Figure H.36 Example output CuneiForm with appropriate instantiation using a simple CuneiForm canvas template example. We defined a Black Swan Scenarios Validation Dataset. Table I.32 CuneiForm Pictorial situation articulation Table I.33 Characteristic Training Classes definitions for a CuneiForm abstract image Figure I.20 Example CuneiForm and instantiated image
[R] : Safety Requirements Justification Report	4	A structured report validating that the decomposed safety	The systems approach artefact must provide traceability and justification for each safety requirement,	H, I.6 Stage 3A: HazTOPs and Ordered AIC-driven Autonomous System Requirements Development

		requirements adequately maintain the intent of the original requirements.	ensuring they correctly address the identified hazards and are properly assigned to system components.	H,I.8 Stage 4: Disordered AIC-driven Black Swan Scenarios Predictions and Validation H,I.10 Stage 6: Black Swan-driven ML Development and Testing H,I.9 Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Epistemic Intelligence Development
[S] : Safety Requirements Argument Pattern [T] : Safety Requirements Argument	4	A structured framework for demonstrating that the safety requirements at each tier adequately capture the intent of the previous tier's requirements.	The systems approach artefact must logically show that the defined safety requirements at each tier align with those from the previous tier, maintaining intent and ensuring completeness.	L.4.5.1 Argument Pattern for Eagle Robot Design Assurance L.4.5.2 Argument Pattern for AVOID system Design Assurance
[V] : AS Development Log	5	A comprehensive record of the design evolution, decisions, and iterations taken throughout development.	The systems approach artefact must track all design changes, safety considerations, and iterations, ensuring full traceability of design decisions.	All STCoTs present a clear argument on the thought processes that went into all engineering judgement.
[W]: tier n Design	4	The design specification at the current tier of decomposition defines system components and	The systems approach artefact must explicitly define design requirements which meet the higher-level safety requirements,	Table H.37 Eagle Drone Safety Training Requirements for Black Swan Scenarios Figure H.36 Example output CuneiForm with

		their interactions.	ensuring that architectural decisions address identified risks and failure modes.	appropriate instantiation using the CuneiForm canvas template example. We defined a Black Swan Scenarios Validation Dataset. Table H.31 4HnWs method for Eagle Drone adjusting patrol functionality Stage 3A, Step 4) Extended Concrete Safety Concept and ML Safety Training Concept Table I.31a AVP Training Requirements for Black Swan Scenarios Table I.31b ML Safety Training Requirements and Perception Dataset Specifications for AVP Figure I.20 Example CuneiForm and instantiated image
[WW] : AS Decision Analysis Report	2	A report analysing the decisions made by the AS at key decision points within different operating scenarios.	The systems approach artefact must demonstrate all decision points have been identified and analysed for potential hazards, considering different environmental conditions and belief states of the AS. Should	H.6.1 Predictive Thinking Pipeline 1: Introducing Autonomous systems into Feed-forward complexity H.6.2 Predictive Thinking Pipeline 2: Designing the affecting Backward-Feed complexity field H.9 Stage 5: CuneiForm-based Syllabus for Safety-

			include examples of decision failures and their potential hazardous outcomes.	Driven ML Epistemic Intelligence Development I.6.1 Predictive Thinking Pipeline 1: Introducing Autonomous systems into Feed-forward complexity I.6.2 Predictive Thinking Pipeline 2: Designing the affecting Backward-Feed complexity field I.9 Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Epistemic Intelligence Development
[X] : Design Process for tier n	5	A structured process outlining the methodology for developing and validating the AS design at tier n.	The systems approach artefact must document and justify the design approach, ensuring that potential hazards are considered and that design decisions align with safety requirements.	L.1 STCoT Validation Report Template L.3 STCoT Validation Report for Unsafe Train Tracks case study Stage (1-6) in AIC Systems Approach H.10.5 Experiment 1: Limited Live video-based experimentation H.10.6 Experiment 2: ML development environment-based validation L.1 STCoT Validation Report Template L.2 STCoT Validation Report for AVOIDDS case study

				Stage n in AIC Systems Approach
[XX]: AS Hazardous Scenarios Definition	2	A comprehensive specification of all identified hazardous scenarios, including the interactions, environment states, and decisions leading to unsafe outcomes.	The systems approach artefact must document hazardous scenarios using the structure: <AS Operating Scenario><Relevant Environment State(s)> AND <Decision>. Should include examples of failure modes, decision errors, and unsafe interactions.	[The research did not include the imposed structure] H.6.3 Predictive Thinking Pipeline 3: Hazards, Threats and Opportunities Scenarios (HazTOPs) Analysis. Figure H.16 Hazards Complexity Field Scope: graphically scoping the hazards within the complexity field by placing hazard icons on target interaction. Figure H.17 Threats Complexity Field Scope Figure H.18 Opportunities Complexity Feels Scope H.6.3.2 Step 2) Characterise the scoped interactions. Figure H.19 Hazards associated with Eagle Drone preventing derailed train complexity field Table H.24 The table describes the AIC interaction dynamics between Eagle Drones and adversarial drones H.6.3.3 Step 3) Apply predictive potential complications guide words.

				Table H.25 HazTOPs Analysis of "3 Drones Attack" Scenario with Risk and Surprise Assessment Table H.26 HazTOPs Analysis of adversarial drone using smart lasers scenario Table H.27 HazTOPs Analysis of adversarial drone hiding behind fence scenario Figure H.20 Soft Hazard Complexity Field Model I.6.3 Predictive Thinking Pipeline 3: Hazards, Threats and Opportunities Scenarios (HazTOPs) Analysis. Figure I.11 Sources of Hazards AIC Complexity Field I.6.3.2 Step 2) Characterise the scoped interactions. Table I.22 Considering Hazards related to I1 interaction I.6.3.3 Step 3) Apply predictive potential complications guide words. Table I.23 Example "More" guide word complication
[Y] : AS Design Justification	5	A structured report providing justification for	The systems approach artefact must explain how design choices	All Predictive Thinking Pipelines and design steps' outputs

		each design decision made at tier n, ensuring alignment with safety requirements.	ensure safety, robustness, fault tolerance, and runtime monitoring while addressing potential risks.	
[YY]: AS Hazardous Scenarios Validation Report	2	A validation document confirming the completeness and correctness of the identified hazardous scenarios.	The systems approach artefact must provide evidence of review, expert validation, simulation-based verification, or real-world testing. It should justify that no significant hazardous scenario has been overlooked.	Validation is done by documenting expert reviews of architect assertions and predictions and the appropriate application of STCoT. A validation report template has been generated, which can be found in L.1 STCoT Validation Report Template Examples: L.2 STCoT Validation Report for AVOIDDS case study L.3 STCoT Validation Report for Unsafe Train Tracks case study
[Z] : AS Design Review	5	Independent reviewers formally assess the AS design to ensure compliance with safety requirements.	The systems approach artefact must verify that design choices do not introduce new hazards, confirm adherence to the design process, and evaluate robustness measures.	Validation is done by documenting expert reviews of architect assertions and predictions and the appropriate application of STCoT. A validation report template has been generated, which can be found in

				L.1 STCoT Validation Report Template L.2 STCoT Validation Report for AVOIDDS case study L.3 STCoT Validation Report for Unsafe Train Tracks case study
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L.5 Eagle Robot AMLAS Safety Case Argumentation Patterns

L.5.1 Stage 1. ML Safety Assurance Scoping

This stage defines the argument for the ML component's boundaries and safety assurance objectives. The AIC Systems Approach Processes involved to satisfy the objectives:

- Stage 1: Uncertainty Problem Articulation and Operational Environment Modelling
- Stage 2: Architect Intent and Autonomous Solution Needs Definition.
- Stage 3A: HazTOPS and Ordered AIC-driven Autonomous System Requirements Development.
- Stage 4: Disordered AIC-Driven Black Swan Scenarios Prediction.

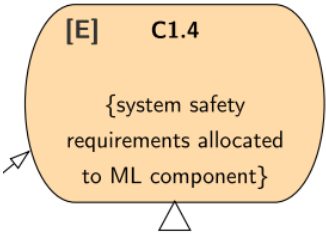
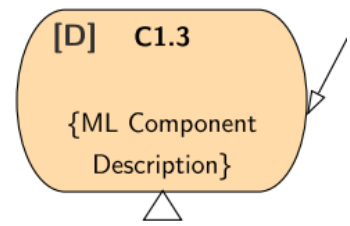
Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts
	H.6 Stage 3A: HazTOPs and Ordered AIC-driven Autonomous System Requirements Development H.8 Stage 4: Disordered AIC-Driven Black Swan Scenario Predictions and Validation
	Table H.34 ML Safety Requirements Derivation Table H.37 Eagle Drone Safety Training Requirements for Black Swan Scenarios
	Table H.15 Architect High-Level Solution Prescription Table H.16 Architect High-Level Solution Prescription related to the impact of roaming adversarial drones

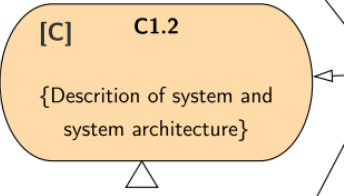
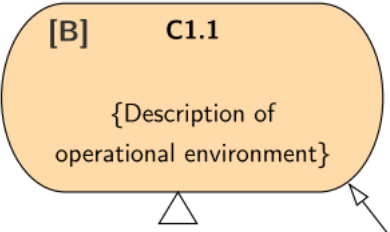
	Table H.17 Architect High-Level Solution Prescription related to the police incapability to capture adversarial drone
	Table H.15 Architect High-Level Solution Prescription Figure H.28 AIC hierarchical modelling schema for 4HnWs analysis Table H.23 The table outlines the interactions and behavioural influences of Eagle Drone Figure H.15 Corrected Eagle Drone complexity field Table H.33 Safety requirements derivations to mitigate the concealed drone problem. Table H.34 ML Safety Requirements Derivation Table H.37 Eagle Drone Safety Training Requirements for Black Swan Scenarios For model architecture: ML Model Type: Roboflow 3.0 Object Detection (Fast)
	H.4.1.5 Architect Prediction 1.5 Table H.12 Police Force Response to Adversarial Drones in Train Track Zones Table H.13 Predicted Factors Output Table H.14 Problem domain factors definitions

	Table H.35 Operational Design Definition for Eagle Robot Deployment in Train Track Zone
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Table L.8 AMLAS Stage 1 artefacts and AIC approach mapping

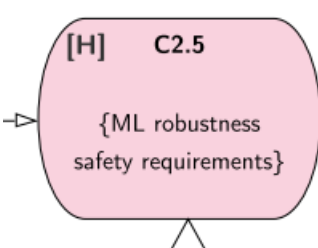
AMLAS Artefact	Explanation	The substantiating AIC methods and artefacts
[A]: System Safety Requirements	The safety requirements define the acceptable risk level and the system's necessary controls.	Stages 3A and 4 clearly capture and analyse the problem domain to derive safety requirements. For example, H.6 Stage 3A: HazTOPs and Ordered AIC-driven Autonomous System Requirements Development
[B]: Description of Operating Environment of System	A formal description of the operational conditions, environmental factors, and constraints in which the ML component will function.	Stage 1 clearly and comprehensively articulates the problem domain defining the operational environment. For example, H.4.1.5 Architect Prediction 1.5.
[C]: System Description	A structured description of the overall system architecture, including interactions between components and the role of ML.	Stage 2 of the AIC process defines the architect's and stakeholders' needs for what the systems should do in response to problematic situations in the problem domain. For example, Table H.15 Architect High-Level Solution Prescription.
[D]: ML Component Description	Defines the function, scope, and interactions of the ML component within the system.	

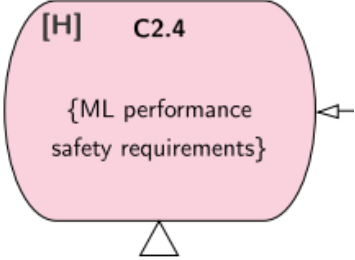
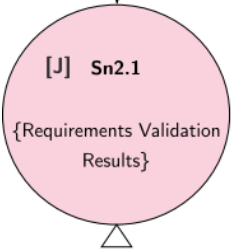
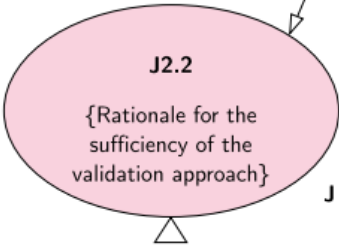
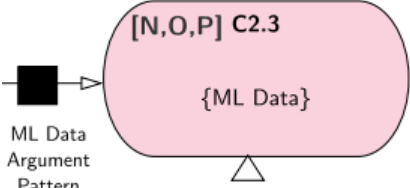
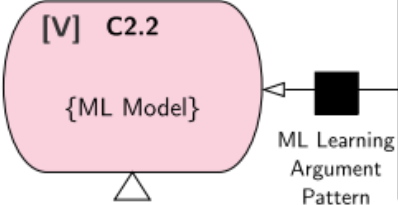
[E]: Safety Requirements Allocated to ML Component	The ML component must meet specific safety requirements to ensure system safety.	The process involved the derivation of ML safety training requirements from system-level safety concepts. For example, Table H.34 ML Safety Requirements Derivation.
[F]: ML Assurance Scoping Argument Pattern [G]: ML Safety Assurance Scoping Argument	A structured argument pattern that ensures the ML component is correctly scoped within the broader safety case.	We captured the argument in the following section: L.5.1 Stage 1. ML Safety Assurance Scoping

L.5.2 Stage 2. ML Safety Requirements Assurance

This stage develops the argument that demonstrates the safety requirements specific to the ML system are defined and validated. The AIC Systems Approach Processes involved to satisfy the objectives:

- STCoT Validation Report for Unsafe Train Tracks case study
- Stage 3A: HazTOPS and Ordered AIC-driven Autonomous System Requirements Development.
- Stage 4: Disordered AIC-Driven Black Swan Scenarios Prediction
- Stage 6: Black Swan-driven ML Development and Testing

Assurance Argument Pattern	AIC Systems Approach: supportive methods and artefacts The following process outputs satisfy the artefact's demonstration requirements:
	Table H.34 ML Safety Requirements Derivation Table H.37 Eagle Drone Safety Training Requirements for Black Swan Scenarios 7.2.2 Step 1: Design CuneiForms

	Table H.40 Characteristic Training Classes definitions for a CuneiForm abstract image 7.8 Chosen training and testing strategies with required performance The table includes the type of robustness (CuneiForms) used for validation and testing. It also includes the minimum performance requirement.
	L.3 STCoT Validation Report for Unsafe Train Tracks case study 7.2.3 Step 2: SOC Compliance 7.2.5 Step 4: CuneiForm Compliance Validation
	L.3 STCoT Validation Report for Unsafe Train Tracks case study Table H.37 Eagle Drone Safety-Training Requirements for Black Swan Scenarios Table H.34A ML Safety Requirements Derivation
	[N] Development Data, [O] Internal Test Data, [P] Verification Data : Dataset Config7 (augmented)
	ML Config7: All in dataset See ML Config7 (augmented)

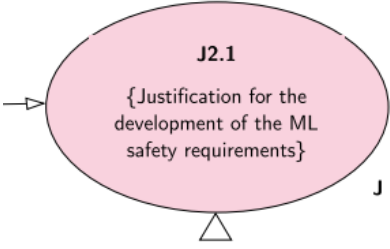
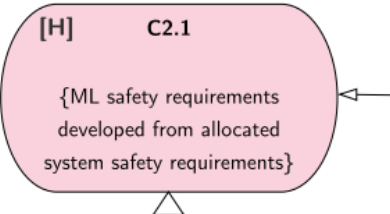
 J2.1 {Justification for the development of the ML safety requirements}	H.6 Stage 3A: HazTOPs and Ordered AIC-driven Autonomous System Requirements Development H.8 Stage 4: Disordered AIC-Driven Black Swan Scenarios Predictions and Validation
 [H] C2.1 {ML safety requirements developed from allocated system safety requirements}	Table H.34 ML Safety Requirements Derivation Table H.37 Eagle Drone Safety Training Requirements for Black Swan Scenarios Table H.38 ML component training dataset requirements

Table L.9 AMLAS Stage 2 artefacts and AIC approach mapping

AMLAS Artefact	Explanation	The substantiating AIC methods and artefacts
[H]: ML Safety Requirements	The ML model must satisfy explicit safety requirements, including performance and robustness constraints. Must demonstrate clear, measurable ML safety constraints (e.g., accuracy, robustness to environmental variations) and how they mitigate system hazards.	The process involved deriving ML safety training requirements from system-level safety concepts. For example, see Table H.34 ML Safety Requirements Derivation.
[I]: ML Safety Requirements Argument Pattern [K]: ML Safety Requirements Argument	Must justify the translation of system-level safety requirements into ML-specific requirements, ensuring relevance, completeness, and correctness.	We captured the pattern in section: L.5.2 Stage 2. ML Safety Requirements Assurance

[J]: ML Safety Requirements Validation Results	The documented results of validation activities performed on ML safety requirements. Must demonstrate that ML safety requirements correctly reflect system safety needs and can be realistically implemented and verified.	In stage 6, for the unsafe training tracks problem domain, we performed the following validation activities: 7.2.3 Step 2: SOC Compliance To validate that CuneiForm correctly captures safety-training requirements. See the following sections in this chapter, which capture the way we would validate CuneiForms tractability to safety requirements: • 7.1.3, 7.1.5 7.2.5 Step 4: CuneiForm Compliance Validation In this step, we performed a validation process to determine whether an image is compliant with a CuneiForm, which is compliant with ML safety training requirements and, in turn, with SOC requirements. Appendix D captures the CuneiForm validation report for AVOID dataset. In addition to direct compliance of CuneiForms, we also provide an audit trail demonstrating how the
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		training syllabus is achieving compliance towards safety requirements preservation during unforeseen (black swan) scenarios. See Table 7.12 Black Swan-driven incremental ML development process to assure model performance under Black Swan operations. We also developed a STCoT validation report template: L.1 STCoT Validation Report Template Which we instantiated to: L.3 STCoT Validation Report for Unsafe Train Tracks case study
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L.5.3 Stage 3. Data Management

This stage develops the argument that data used in training and validation meets the necessary quality and safety standards. The AIC Systems Approach Processes involved to satisfy the objectives:

- Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Epistemic Intelligence Development
- CuneiForm Training syllabus as a Validation Process for Datasets
- Stage 6: Black Swan-driven ML Development and Testing

The following dataset validation artefacts can be used as evidence to demonstrate how the dataset captures a CuneiForm, which in turn shows how a safety requirement is captured in a dataset (example taken from AVOIDDS case study):

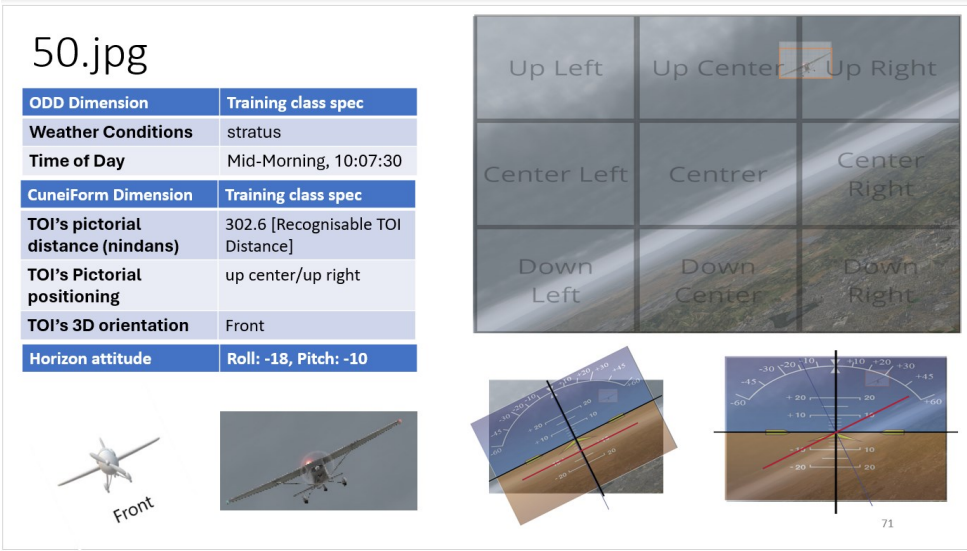
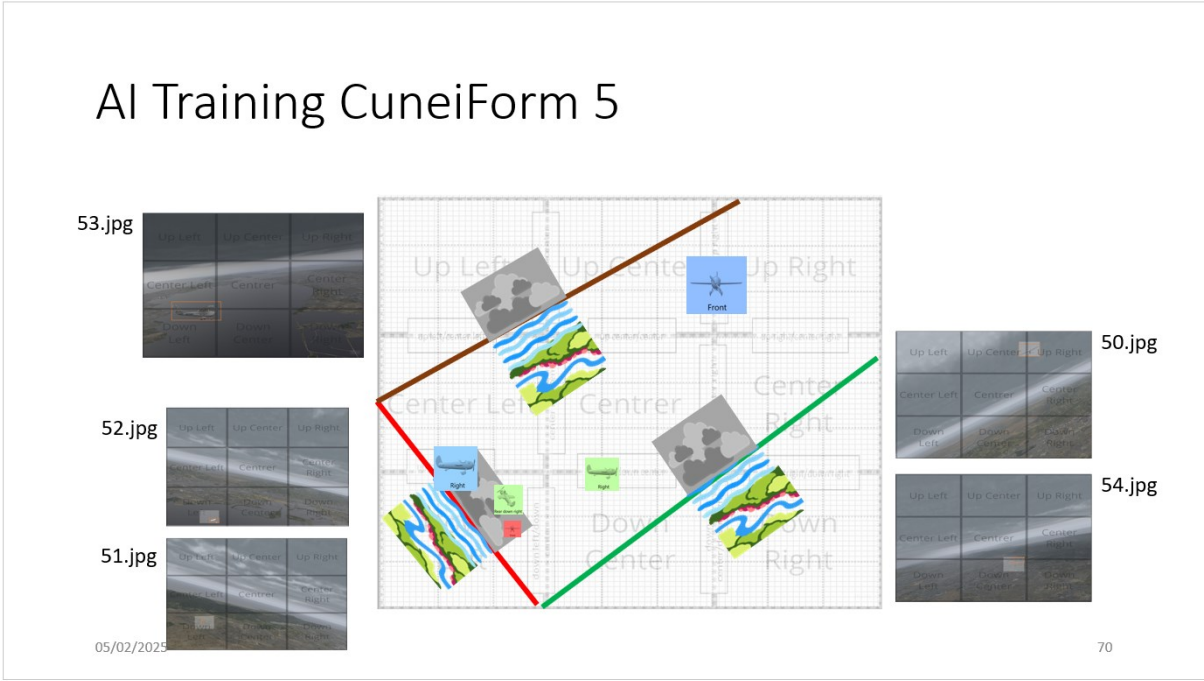


Figure L.2 Example validation report of how a sample training image validates CuneiForm 5

CuneiForm 5 has been retrospectively generated:

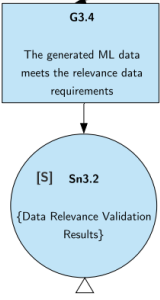
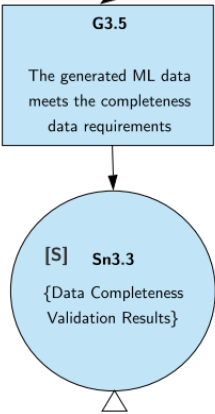
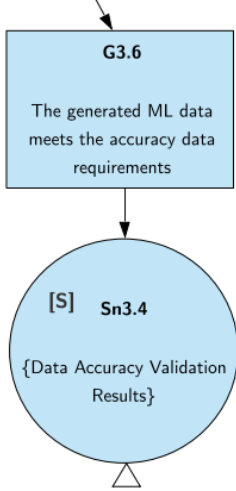


[L]: Data Requirements	Defines the ML data's characteristics to ensure the model meets its safety requirements. It must include data relevance, completeness, accuracy, and balance requirements, ensuring sufficient operational domain representation.	The AIC systems approach included a process for systematically identifying data requirements derived from the training concept. For example, Table H.38 ML component training dataset requirements Table 6.35 Black Swan Scenarios Batch A and B CuneiForms Table 6.36 Typical operations CuneiForms Figure 6.36 H.54 Out-of-context CuneiForm of drones Figure 6.27 represents the Cuneiform (H.36) characterisation for the Black Swan Scenario Validation. Figure 7.20 Example CuneiForm and instantiated image for AVOIDDS case study
[M]: Data Requirements Justification Report	Justifies that the data requirements are sufficient to develop a safe ML model. It must demonstrate how the data requirements were derived, validated, and justified to capture the necessary variations in the operational environment.	The following describes the systematic process that demonstrates how the data requirements were derived: I, H.9 Stage 5: CuneiForm-based Safety-Driven ML Training, Testing Process

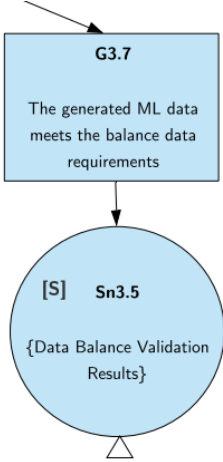
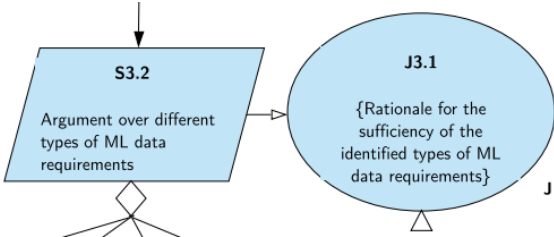
		I.10, Section 7.2 Stage 6: Dataset generation and validation process
[N]: Development Data	Data used for training and validating the ML model during development.	We developed a dataset using the requirements derived from stage 4 and 5. We also used a pre-existing dataset and injected the specially derived dataset and then validated its performance against a Black Swan scenario. For more details, see section L.5 and the following link to the output model . [N] Development Data, [O] Internal Test Data, [P] Verification Data : Dataset Config7 (augmented)
[O]: Internal Test Data	Data used for testing the ML model internally before verification. This is equivalent to what we define as a validation dataset.	
[P]: Verification Data	A separate dataset, equivalent to what we define as a test dataset, is used for final model evaluation to assess performance under unseen conditions.	
[Q]: Data Generation Log	Documents the process of data collection, pre-processing, and augmentation. It must capture decisions made during data collection, processing, and augmentation, providing a rationale for data sufficiency.	The CuneiForm method in stage 5 defines the rationale of how data requirements are translated into the dataset. Also, 7.2.2 Step 1: Design CuneiForms, and 7.2.4 Step 3: Instantiates CuneiForms
[S]: ML Data Validation Results	Documents the validation outcomes, ensuring the generated data meets ML data requirements. It must demonstrate that data relevance, completeness, balance, and accuracy were	In the unsafe train tracks case study, we did not consider the validation report over the produced dataset. However, in the AVOIDDS case study, we did. Therefore, we will include

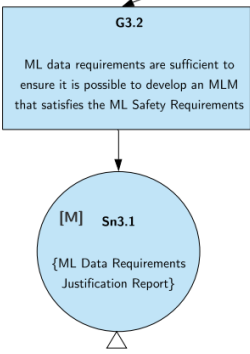
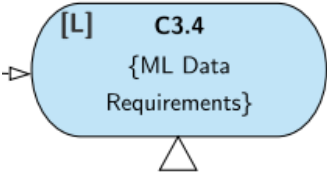
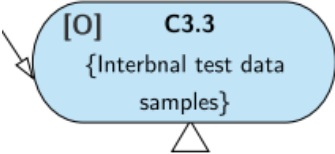
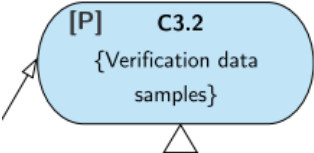
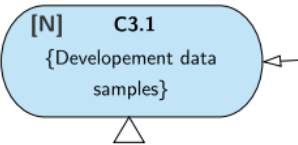
	verified and any discrepancies justified.	the following from the second case study: I.10.1 CuneiForm Training syllabus as a Validation Process for Datasets I.10.3 CuneiForm Validation Artifact Safety Validation Report for AVOID Dataset using the CuneiForm method [can be retrieved from here] As for unsafe train tracks, an example CuneiForm validation can be found in section 7.2. Specifically see section 7.2.5 Step 4: CuneiForm Compliance Validation
[R]: ML Data Argument Pattern [T]: ML Data Argument	A structured assurance argument justifying the adequacy of the ML data for safety assurance.	We captured the artefacts in the following sections: L.5.3 Stage 3. Data Management

Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts The following process outputs satisfy the artefact's demonstration requirements:
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	H.9 Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Epistemic Intelligence Development An example CuneiForm validation can be found in section 7.2. Specifically see section 7.2.5 Step 4: CuneiForm Compliance Validation
	H.9 Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Epistemic Intelligence Development An example CuneiForm validation can be found in section 7.2. Specifically see section 7.2.5 Step 4: CuneiForm Compliance Validation
 Data Accuracy³	7.2.3 Step 2: SOC Compliance Validate that CuneiForm correctly captures safety-training requirements. See the following sections in this chapter, which capture the way we would validate CuneiForms tractability to safety requirements: sections 6.7.3, 6.7.5. Also, 7.2.5 Step 4: CuneiForm Compliance Validation where we validate that data accurately captures the CuneiForms.

³ Data accuracy refers to how reliably a dataset reflects real-world conditions, ensuring labels and metadata truthfully represent the objects or scenarios they describe. For drone detection systems, this means training data (images, sensor readings, labels) must precisely capture drones in diverse, realistic settings to avoid biases or errors that could compromise the model's performance.

	In the AVOIDDS case study, we did. Therefore, we will include the following from the second case study: I.10.1 CuneiForm Training Strategy as a Validation Process for Datasets I.10.3 CuneiForm Validation Artifact Safety Validation Report for AVOID Dataset using the CuneiForm method [can be retrieved from here]
 <pre> graph TD G3.7[G3.7 The generated ML data meets the balance data requirements] --> Sn3.5(([S] Sn3.5 {Data Balance Validation Results})) style G3.7 fill:#add8e6,stroke:#333,stroke-width:1px style Sn3.5 fill:#add8e6,stroke:#333,stroke-width:1px </pre>	In the unsafe train tracks case study, we did not consider the validation report over the produced dataset. However, in the AVOIDDS case study, we did. Therefore, we will include the following from the second case study: I.10.1 CuneiForm Training Strategy as a Validation Process for Datasets I.10.3 CuneiForm Validation Artifact Safety Validation Report for AVOID Dataset using the CuneiForm method [can be retrieved from here]
 <pre> graph LR S3.2[/S3.2/
Argument over different
types of ML data
requirements/] --> J3.1((J3.1
{Rationale for the
sufficiency of the
identified types of ML
data requirements})) style S3.2 fill:#add8e6,stroke:#333,stroke-width:1px style J3.1 fill:#add8e6,stroke:#333,stroke-width:1px </pre>	H.9 Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Epistemic Intelligence Development H.10 Stage 6: Black Swan-driven ML Development and Testing

	H.9 Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Epistemic Intelligence Development H.10 Stage 6: Black Swan-driven ML Development and Testing
	Table H.38 ML component training dataset requirements Table 6.35 Black Swan Scenarios Batch A and B CuneiForms Table 6.36 Typical operations CuneiForms Figure 6.36 H.54 Out-of-context CuneiForm of drones
	[N] Development Data, [O] Internal Test Data, [P] Verification Data : Dataset Config7 (augmented)
	
	

L.5.4 Stage 4. Model Learning

This stage develops the argument for the creation and evaluation of the ML model correctly and comprehensively to ensure safety objectives are achieved. The AIC Systems Approach Processes involved to satisfy the objectives:

- Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Epistemic Intelligence Development
- Stage 6: Black Swan-driven ML Development and Testing

The model training process we produced is characterised below⁴:

Table L.11 Final training syllabus and trained output model

ML Config7: All in dataset	Total Training: 16788 (71%) (no augm.)	Total Valid: 4426 (19%) (no augm.)	Total Test: 2319 (10%) (no augm.)
See ML Config7 (augmented)	Out-of-context images: 10954 (65%)	Out-of-context images: 1969 (44%)	Out-of-context images: 268 (11%)
Total: 23533 images (without augmentation)	Black Swans A CuneiForm Scenarios: H.36, 37, 38, 39, 40, 41: 1866 (11%)	Black Swans A CuneiForm Scenarios: H.36, 37, 38, 39, 40, 41: 622 (14%)	Black Swans A CuneiForm Scenarios: H.36, 37, 38, 39, 40, 41: 622 (26%)
With augmentation: 40321 images	Black Swans B CuneiForm Scenarios: H.51,52,53: 695 (4%)	Black Swans B CuneiForm Scenarios: H.51,52,53: 231 (5%)	Black Swans B CuneiForm Scenarios: H.51,52,53: 232 (10%)
	Typical Operations CuneiForm Scenarios: H.42, 43,44: 3273 (19%)	Typical Operations CuneiForm Scenarios: H.45,46,47: 1604 (36%)	Typical Operations CuneiForm Scenarios: 48,49,50: 1197 (51%)
	Applied Pre-processing: Grayscale: Applied		
	Applied Pre-processing:		

⁴ See section H.9.5 for more details on CuneiForms.

	Outputs per training example: 2 Noise: Up to 2.39% of pixels Performance: mAP50/test 99%, mAP50/validation 98%,
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Table L.12 captures the primary artefacts that need to be presented:

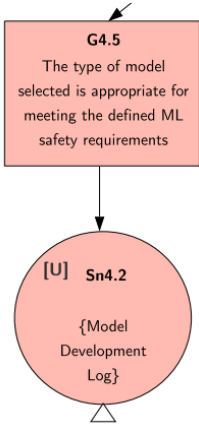
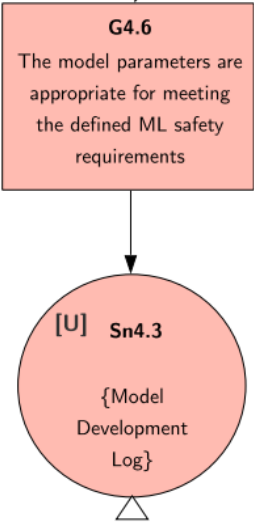
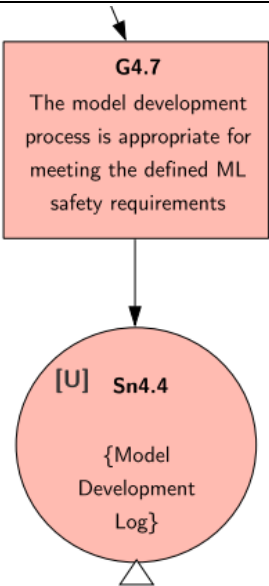
Table L.12 AMLAS Stage 4 artefacts and AIC approach mapping

AMLAS Artefact	Explanation	The substantiating AIC methods and artefacts
[U]: Model Development Log	A record of all decisions, configurations, and justifications during model training and development. It must capture all design choices, hyperparameter settings, and rationale for selecting the final model, document methods to prevent overfitting and ensure robustness.	For ML development audit trail see Table 7.12 Black Swan-driven incremental ML development process to assure model performance under Black Swan operations, Section 7.2.8. We used the RoboFlow platform to capture the model development log activities. The following model is the output of experiment 9.6 where we injected the black swan training subset and validated it against the Black Swan Validation subset: ML Config7 (augmented) Trained based on: Experiment 9.6 full dataset including black swans, typical operations and random out-of-context: [N] Development Data, [O] Internal Test Data, [P] Verification

		Data : Dataset Config7 (augmented) Furthermore, the following stages also capture activities that were directly involved development of the model: H.9 Stage 5: CuneiForm-based Safety-Driven ML Training, Testing Process H.10.6 Captures a comprehensive experimentation to justify the training syllabus and model.
[V]: ML Model	The final ML model was developed based on the training process. It must demonstrate that the trained model satisfies all ML safety requirements, including performance, robustness, and failure handling.	We used the RoboFlow platform to capture the model development log activities. The following model is the output of experiment 9.6 where we injected the black swan training subset and validated it against the Black Swan Validation subset: ML Config7 (augmented)
[X]: Internal Test Results	Documented results of evaluating the ML model on internal test data.	The test dataset can be found here: [N] Development Data, [O] Internal Test Data, [P] Verification Data : Dataset Config7 (augmented) The validation and test results can be found in ML Config7 (augmented)

		Also, Black Swan (only) testing can be found in the following ML configuration 6 which demonstrates the model's ability to handle out-of-distribution, data shift.
[Y]: ML Learning Argument [W]: ML Learning Argument Pattern	The instantiated ML learning assurance argument is based on development and testing evidence.	The instantiated argument is captured in the following section: L.5.4 Stage 4. Model Learning

Assurance Argument Pattern	AIC Systems Approach supportive methods and artefacts The following process outputs satisfy the artefact's demonstration requirements:
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 G4.5 The type of model selected is appropriate for meeting the defined ML safety requirements [U] Sn4.2 {Model Development Log}	For ML development audit trail see Table 7.12 Black Swan-driven incremental ML development process to assure model performance under Black Swan operations, Section 7.2.8.
 G4.6 The model parameters are appropriate for meeting the defined ML safety requirements [U] Sn4.3 {Model Development Log}	
 G4.7 The model development process is appropriate for meeting the defined ML safety requirements [U] Sn4.4 {Model Development Log}	

	The validation and test results can be found in ML Config7 (augmented) Also, Black Swan (only) testing can be found in the following ML configuration 6 which demonstrates the model's ability to handle out-of-distribution, data shift.
	The validation and test dataset can be found here: Dataset Config7 (augmented)
	H.9 Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Epistemic Intelligence Development 7.8 Chosen training and testing strategies with required performance 7.2.5 Step 4: CuneiForm Compliance Validation
	7.2 Stage 6: Black Swan-driven ML Development and Testing
	ML Config7 (augmented)

L.5.5 Stage 5. Model Verification

Outside the PhD scope

L.5.6 Stage 6. Model Deployment

Outside the PhD scope